

Why an Incorrect Identification of Exogenous Consolidation Episodes may affect the Measurement of the Interactions between Monetary and Fiscal Policies: A Case Study

Abstract

This paper examines the interaction between monetary conditions and fiscal consolidation. While lower interest rates may reduce the contractionary costs of fiscal tightening, existing evidence remains inconclusive as to whether easier monetary conditions make consolidations more likely to start, continue, or succeed. This uncertainty is partly related to how fiscal consolidation episodes are identified, since the cyclically-adjusted primary balance (CAPB) - a standard way to measure consolidation episodes - and narrative methods may capture different policy events. The paper asks whether monetary conditions affect fiscal consolidation outcomes, and whether the answer depends on the identification method used. To address this question, it applies a common econometric framework to 16 OECD countries over 1978–2014, comparing CAPB-based episodes with narratively identified exogenous consolidation plans. Monetary conditions are measured through changes in short- and long-term interest rates, including a Taylor-rule based indicator, while controlling for fiscal pressure, cyclical conditions, exchange-rate movements, the composition of the fiscal adjustment, and euro-area institutional regimes. The results show limited and mixed evidence that monetary easing improves consolidation outcomes. Monetary variables matter in some CAPB-based specifications, but the relationship weakens under narrative identification. Overall, fiscal needs, macroeconomic conditions, institutions, and the composition of the fiscal adjustment appear more robust than the degree of monetary accommodation in explaining consolidation outcomes.

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"Endings are just bookmarks for new beginnings"

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1 Introduction

Whether easier monetary conditions* can make fiscal consolidation less costly, more durable and ultimately more successful is a central question in the study of monetary-fiscal interactions. This paper argues that the answer depends not only on the stance of monetary policy, but most importantly on how fiscal consolidation itself is identified and subsequently analysed. Before assessing whether lower interest rates support fiscal adjustment, it is first necessary to establish whether the observed episode is the result of an exogenous consolidation policy decision, rather than an improvement in the fiscal balance generated by changes in macroeconomic conditions. This paper is thus concerned with two closely related questions: how monetary conditions affect a fiscal consolidation, and whether an incorrect identification of exogenous consolidation episodes can distort the measured interaction between monetary and fiscal policies.

The theoretical relationship between monetary conditions and fiscal consolidation is ambiguous. On the one hand, lower interest rates may facilitate fiscal retrenchment by offsetting part of the contractionary effect of austerity, supporting aggregate demand and reducing the interest burden on public debt. On the other hand, easier monetary conditions may weaken the pressure to consolidate by improving the government's budget constraint, while falling long-term rates may reflect market confidence in a credible fiscal package rather than an independent monetary stimulus. Moreover, central bank independence limits the extent to which monetary authorities can be expected to react directly to fiscal policy decisions. For this reason, the observed association between interest rates and fiscal adjustment cannot be interpreted mechanically as evidence that monetary policy causes consolidations to begin, continue or succeed.

This ambiguity is particularly important because much of the literature identifies consolidation episodes through changes in the cyclically-adjusted primary balance

*In this paper, monetary conditions are measured through interest-rate-based indicators: changes in nominal and real short-term and long-term rates, together with a Taylor-rule-based measure of the monetary stance. The analysis therefore focuses on conventional interest-rate channels of monetary accommodation and does not directly capture other forms of monetary easing that may become relevant when policy rates approach the lower bound.

(CAPB). The CAPB approach is useful because it attempts to remove the direct effect of the business cycle from the fiscal balance, where large improvements in the CAPB are often interpreted as evidence of discretionary fiscal tightening. However, cyclically-adjusted balances may still capture endogenous changes linked to economic conditions, revenue windfalls, forecast errors, interest-rate movements or policy measures adopted in response to the cycle. Therefore, an outcome-based improvement in the CAPB may not always correspond to an exogenous fiscal consolidation. The fiscal response of the Obama administration during the 2008–09 financial crisis illustrates this limitation. While the recession automatically increased government spending and reduced tax revenues, the administration also introduced additional discretionary stimulus in response to the downturn. Such policy measures are not removed by cyclical adjustment and therefore imply that CAPB-based indicators may still capture endogenous fiscal responses to output shocks rather than exogenous fiscal policy shifts (Alesina, Favero, et al., 2019a). By contrast, the narrative approach seeks to identify fiscal measures motivated by long-term deficit reduction rather than short-term stabilization, and is therefore closer to policy intent. The central contribution of this paper is to show that the choice between these two identification methods is not a technical detail, but a key determinant of the estimated monetary-fiscal relationship itself.

The objective of the paper is to assess this issue by applying a common empirical framework to two different definitions of fiscal consolidation. First, episodes are identified using a CAPB-based rule, following the approach of Ahrend et al. (2006), whereby a consolidation begins when the CAPB increases by at least one percentage point of GDP in one year or over two consecutive years. Second, the same analysis is repeated using narratively identified exogenous fiscal adjustment plans drawn from the Alesina et al. database (IGIER, 2026), which builds on the narrative identification strategy of Pescatori et al. (2011). The purpose is therefore not only to estimate whether monetary conditions matter, but to evaluate whether the answer changes when the treatment variable is defined as an observed fiscal-balance improvement rather than as a documented policy action.

The analysis focuses on 16 OECD countries between 1978 and 2014 and examines

a fiscal consolidation across three phases: the start of the episode, its continuation, and its success. Monetary conditions are measured using changes in nominal and real short-term and long-term interest rates, as well as the change in the short-term rate relative to a Taylor-rule benchmark. These monetary variables are evaluated alongside controls for the initial need for fiscal adjustment, the cyclical position of the economy, the real effective exchange rate and the composition of the fiscal packages. Particular attention is also given to credibility and reverse causality, since interest rates may themselves respond to the perceived quality of fiscal adjustment. For this reason, the analysis controls for the composition of consolidation packages. Following Ahrend et al. (2006), it distinguishes between current expenditure cuts and capital expenditure cuts. Current expenditure refers to recurrent government spending, such as public wages, purchases of goods and services, transfers, subsidies and pensions. Capital expenditure, instead, refers mainly to public investment, including infrastructure and other spending that contributes to the public capital stock. This distinction is useful because different types of expenditure cuts may convey different information about the credibility and quality of the adjustment. However, it is interpreted as an imperfect proxy rather than as a direct measure of durability. Current spending cuts are not automatically more credible, and capital spending cuts are not necessarily easier to reverse. More broadly, consistent with Alesina, Favero, et al. (2019b), credibility depends on whether the measures are perceived as structural, persistent and politically costly to implement. Reforms to pensions, transfers or public-sector wages may therefore signal commitment more strongly than the simple current-versus-capital classification would suggest. This matters because credible expenditure-based consolidations may themselves lead to lower interest rates, making it necessary to control for composition when assessing whether monetary conditions independently support fiscal consolidation.

The main findings are striking. Although the CAPB and narrative methods identify a similar number of consolidation episodes, they do not identify the same treatment periods. CAPB-based episodes are often more continuous, while narrative episodes tend to be shorter and more fragmented, reflecting the difference between fiscal-balance outcomes and documented policy decisions. This matters for the interpretation of monetary

variables. Under CAPB-based identification, there is some evidence that easier-than-rule monetary conditions are associated with the start of consolidation, and that falling short-term rates improve selected debt-dynamic outcomes. However, these effects do not extend consistently to continuation, cumulative adjustment, episode length or post-episode durability. Under narrative identification, the evidence for a supportive monetary channel is much weaker: short-rate easing is associated with the start of exogenous consolidation episodes, but monetary variables are generally insignificant for continuation and for most measures of success. In some specifications, the signs are even inconsistent with a simple monetary-accommodation interpretation. Thus, monetary conditions may shape the environment in which fiscal consolidation takes place, especially in the initial phase, but they do not emerge in this analysis as a stable and general determinant of whether consolidation is continued or successful. Instead, the more robust determinants are fiscal need, weak macroeconomic conditions, institutional constraints and the credibility and composition of the adjustment package.

The insights from these findings are therefore methodological as well as substantive: the estimated role of monetary policy depends on whether consolidation is identified as an observed improvement in the fiscal balance or as an exogenous policy decision. In this sense, incorrect identification can make monetary policy appear more important than it is, because the estimated monetary channel may partly capture endogenous fiscal-balance movements, credibility effects or common responses to macroeconomic conditions.

The paper is structured as follows. Section 2 reviews the theoretical and empirical literature on monetary-fiscal interactions during fiscal consolidation episodes. It first discusses the mechanisms through which monetary conditions may affect the start, continuation and success of fiscal adjustment, before turning to the distinction between CAPB-based and narrative identification. Section 3 presents the data and method of analysis. It describes the construction of the CAPB- and narrative-based episode samples, the monetary-stance indicators, the control variables and the econometric specifications used in the analysis. Section 4 compares the consolidation episodes identified under the two approaches and summarises the main econometric results. Section 5 discusses these re-

sults in detail, distinguishing between the start, continuation and success of consolidation episodes, before assessing what the comparison between identification methods implies for the interpretation of the monetary-fiscal relationship. It then considers the limitations of the study and possible directions for future research. Section 6 concludes by evaluating whether monetary conditions can be interpreted as an independent and stable determinant of fiscal consolidation.

2 Theoretical Background

2.1 Interactions between Monetary and Fiscal Policies

The interaction between monetary policy and fiscal policy with regards to the effect on consolidation episodes can be examined from three main angles: their start, continuation and success. Although definitions vary across studies (Yang et al., 2015), amongst them there is no clear, supportive evidence on the effect of monetary policy on fiscal consolidations from either point of view. Clearly, a key factor to consider is central bank independence which explicitly prevents direct responses to fiscal interventions. Nonetheless, there is some acknowledgement that austerity policies are rarely implemented in isolation because by lowering interest rates and buying government bonds, the central bank can help (Alesina, Favero, et al., 2019b). Yet, Ardagna (2004), amongst others, concludes that successful fiscal contractions are not the result of accompanying monetary policy. Similarly, Von Hagen et al. (2001) conclude that although monetary easing may increase the probability that a fiscal consolidation will be pursued, it has no implications on the likelihood of its success. More specifically, they conclude that once only successful consolidation episodes are considered, the impact of monetary policy becomes irrelevant. The causal path therefore remains unclear as lower interest rates may offset the contractionary effects of austerity but they may also ease a government's budget constraint by stimulating revenue growth and reducing interest payments on public debt, thereby weakening the pressure to consolidate.

However, a notable exception to these generally inconclusive findings is the paper by Ahrend et al. (2006). Whilst they acknowledge that the extent to which fiscal consolidation can be assisted by monetary easing is heavily constrained by economic circumstances, they conclude that a supportive monetary policy stance can help. By undergoing multivariate analysis, controlling for macroeconomic indicators that indicate a crisis situation, they find that consolidation efforts are more likely to be pursued and to succeed if the monetary policy stance is eased in the initial stages of the episode, thereby offsetting the potentially adverse impacts of austerity. Further, to address issues of reverse causality, by establishing, and subsequently controlling for, an indirect reaction of monetary policy and financial markets to the credibility of austerity (short and long term interest rates are more likely to fall during episodes characterised by a greater reliance on expenditure cuts), they are still able to conclude, albeit to a weaker extent, that changes in monetary stance do positively affect the chances that a fiscal consolidation episode will be successfully pursued.

In support of this, a later paper by Molnár (2012) finds that declining interest rates do increase the probability of success and duration of consolidation episodes, despite them not having a robust impact on the start, size or intensity. Alesina and Ardagna (2013) support this in part by suggesting that a reduction in long-term interest rates may either be associated with an increase in confidence or it may be the result of monetary easing (endogenous to the fiscal adjustment) because of the perceived credibility of a given fiscal consolidation policy package, thus finding that easier monetary conditions appear to have helped adjustments to be successful.

Regardless, there is however, consensus and strong empirical evidence that the pursuit, success and impact of fiscal consolidation episodes depends on their credibility, and consequently their composition. Most notably, tax based austerity plans often lead to deep and prolonged recessions, whereas expenditure based plans have a temporary contractionary effect that dissipates on average within two years after its introduction (Alesina, Favero, et al., 2019b). Further, with regards to starting episodes of fiscal consolidation, these tend to occur when large budget deficits threaten sustainability and

when other macroeconomic indicators suggest a crisis situation, which implies that fiscal consolidations may not be distributed evenly across the business cycle (Drazen and Grilli, 1990). Indeed, Ahrend et al. (2006) motivate this by suggesting that the political consensus needed to undertake austerity may be easier to obtain in a situation of crisis.

2.2 Methods of Identification

With regards to the empirical identification of fiscal consolidation episodes, there are a variety of methodologies prevalent in the literature. The majority of the studies that focus directly on investigating the effect of monetary conditions on fiscal consolidation use the cyclically adjusted primary balance to identify fiscal consolidation episodes by establishing thresholds of reductions in the CAPB as a percentage of GDP in a given period of time. This is consistent with many other related fiscal macroeconomic studies, as highlighted by Yang et al. (2015) and Molnár (2012). The motivation for using the CAPB as a tool is that it aims to mute the reverse effects of the business cycle on the government balance by theoretically producing the deficit that a country would have if the economy ran at full potential. The rationale for this approach is that a year with a large reduction of the primary deficit cannot be considered to be “business as usual” and must therefore represent a change in fiscal policy (Alesina and Ardagna, 2013). However this is likely to suffer from endogeneity, as even if the change in primary balance is cyclically adjusted, it still includes other legislated changes in taxes and spending that are motivated by the state of the economy (Alesina, Favero, et al., 2019b). Furthermore, Yang et al. (2015) outlines that a generic threshold for a change in CAPB fails to consider the idiosyncrasy of fiscal policy in each country and therefore a better approach would be to incorporate size, persistence and country-specific heterogeneity in CAPB definitions. Overall, using the CAPB to identify episodes of fiscal consolidation is problematic as adjustment methods suffer from measurement errors that are likely to be correlated with economic developments, making changes in the CAPB correlated with economic activity but not necessarily to policy actions. More importantly, even if the change in CAPB

can accurately reflect discretionary changes in fiscal policy, these may be induced by an intention to respond to cyclical fluctuations (Pescatori et al., 2011).

Therefore, to address this endogeneity, the narrative method, originally introduced by Romer and Romer (2004) and subsequently extended and applied throughout the literature, uses official documents motivating policy measures to select shocks not aimed at stabilizing the cycle. Unsurprisingly, the difference between cyclically adjusted shocks and narratively identified ones can be sizable (Alesina, Favero, et al., 2019b). The way these time series datasets are built varies slightly across papers, however, for example Alesina, Favero, et al. (2018) amplify a set of narratively identified exogenous fiscal consolidation episodes from Pescatori et al. (2011) by collecting additional information on fiscal measures across OECD countries and then extending it by another five years to cover the full period between 1978 up to 2014.

Ultimately, as the majority of studies employing narrative methods focus broadly on measuring the effects of fiscal policy, it is of interest to use such tools to investigate the effects of monetary conditions on fiscal consolidations, and determine how these results compare to the conclusions reached by Ahrend et al. (2006), for example.

3 Methods

3.1 Data Collection

In order to effectively establish whether monetary conditions affect fiscal consolidation, and whether this differs based on how episodes are identified, this paper will conduct the same analysis twice, once using episodes identified using the CAPB and once using episodes identified using the narrative approach across an otherwise identical dataset.

The baseline dataset used is the one constructed by A. Alesina, G. Azzalini, O. Barbiero, C. Favero, F. Giavazzi, M. Karamysheva, A. Miano and M. Paradisi, which is publicly available via the IGER website (IGIER, 2026). This dataset contains yearly exogenous fiscal adjustment plans for 16 OECD countries (Australia, Austria, Belgium,

Canada, Denmark, Finland, France, Germany, Ireland, Italy, Japan, Portugal, Spain, Sweden, United Kingdom, United States) between 1978 and 2014. It builds on the narrative identification procedure of Pescatori et al. (2011), but groups the measures by the years in which they were introduced, instead of grouping by the year of effective implementation. This database also includes a variety of macro variables mostly obtained from the OECD. To this dataset, the cyclically adjusted primary and overall balance, output gap and the nominal short- and long- term interest rates are added directly from the OECD (OECD, 2026).

Further, OECD Medium-Term Baseline Projections are also included. These projections are needed to construct the “need for adjustment” synthetic indicator used by Ahrend et al. (2006) to control for initial levels of the overall budget balance and public debt as a percentage of GDP before a consolidation episode begins.

3.2 Method of Analysis

The method of analysis will follow closely that put forward by Ahrend et al. (2006), with some modifications. The interaction of monetary policy with episodes of fiscal consolidation will be evaluated by considering their commencement*, continuation and success, whilst controlling for the influence of other plausible determinants of monetary and fiscal policy in a multivariate econometric setting.

3.2.1 Defining Episodes and Indicators

To construct the dataset, the narrative and CAPB analyses differ only in the set of fiscal consolidation episodes initially identified. All subsequent variables are then constructed using the same procedure, but are defined with reference to the consolidation episodes identified under each method. Specifically, to identify episodes of fiscal consolidation using the CAPB, for consistency, the definition used by Ahrend et al. (2006) is

*Due to the constraints of the datasets available, in the case of CAPB-based identified episodes, the commencement refers to the observed improvement in the CAPB, whereas for narrative-based identified episodes, commencement refers to the introduction (not implementation) of fiscal consolidation policies.

applied. Therefore, “episodes of fiscal consolidation are identified as starting when the CAPB increases by at least one percentage point of GDP (in one year or over two consecutive years with at least $\frac{1}{2}$ point in the first year)”. The reason for using this definition, which is lower than the common threshold, is because it also allows the analysis of the factors that cause an episode to be sustained or be aborted quickly. For the narrative approach, episode starts are identified using the database dummy variable “New Plan”, which is equal to one in years when a new exogenous fiscal consolidation plan is introduced (IGIER, 2026). Building on these two respective dummies, an episode is then considered to continue for as long as the CAPB continues to increase (with an additional condition imposed for the narrative episodes that ensures that if a new exogenous fiscal measure is introduced whilst the CAPB is increasing, a new episode is considered as starting and is therefore not considered as the continuation of a previous exogenous episode). Further, an episode is then classified as being “seriously pursued” if in the two years following its start, an additional adjustment of at least one percentage point of GDP is achieved.

3.2.2 Establishing Proxies of Monetary Stance

To proxy for the interaction between monetary policy and fiscal policy during fiscal consolidation episodes, five different interest-rate variables are considered. Most directly, the real short-term rate is used as the main proxy for the monetary stance, following Ahrend et al. (2006), and is constructed by deflating the nominal short-term rate with core inflation. In principle, expected inflation would provide a more accurate measure of the ex ante real interest rate relevant for private-sector decisions. However, consistent and comparable measures of expected inflation are not available for all countries and years in the sample. Core inflation is therefore used as a practical proxy to preserve cross-country and time-series comparability. Additionally, nominal short- and long-term rates, as well as nominal and real long-term rates, are considered to capture possible transmission through the term structure of interest rates. Finally, for the purpose of this analysis, it is also relevant to evaluate monetary policy actions that are specifically

above those required for short-term economic stabilization. For this, the change in the monetary policy stance relative to what would be expected to occur under a Taylor rule (with 0.5% weight to changes in inflation and output gap respectively) is used as a proxy. For example, as posited by Ahrend et al. (2006), the monetary stance is likely to be tighter than suggested by a Taylor rule whenever debt sustainability concerns influence inflation expectations either through risks of monetisation or problems with the current account.

Of course, the interpretation of these proxies remains in part ambiguous. For example, lower interest rates may support austerity by offsetting its potentially contractionary impact, but may also discourage it if lower rates are perceived as reducing the need for fiscal adjustment (either through increased revenues or a lower interest burden). Additionally, given that market rates are determined in part by expectations, an observed decrease in long term rates may suggest that the market is reacting to a credible fiscal adjustment plan. Therefore, to avoid these confounding effects, all monetary stance proxies are lagged in the multivariate analysis.

3.2.3 Defining Control Variables

Across all regression specifications, a few critical explanatory variables are included. First, a synthetic indicator representing the “need for adjustment” is constructed using the same definition from Ahrend et al. (2006), where it is built as the difference between the current cyclically-adjusted overall balance and a target level (that required to bring the gross debt ratio from the current level to zero over thirty years under the assumption of nominal income growth in line with actual realizations extended with OECD Medium-Term Baseline projections). With this, the expectation is that adjustments are more likely to commence, continue and succeed when the “need for adjustment” is stronger.

Further, a control for the state of the economy is included via the output gap, indicating the cyclical position of each fiscal adjustment episode. This variable is lagged to avoid simultaneity, and thus shows whether the episode is predominantly pro- or counter-cyclical (if it was considered in level, it would have indicated whether the adjustment took

place in “good” or “bad” times).

Moreover, the lagged real effective exchange rate is also implemented as a control variable as there is some evidence (although not clearly conclusive) that a depreciated currency could facilitate fiscal adjustments by offsetting some adverse contractionary effects, or it may also be an indicator of crisis leading to increased pressure for fiscal consolidation (Ahrend et al., 2006).

3.2.3.1 Addressing Credibility and Reverse Causality

To comprehensively address potential issues of reverse causality, it is important to also control for the quality of adjustment episodes, as episodes that are perceived as credible may themselves cause rates to fall.

To determine whether this is indeed the case, Figures 1 and 2 allow the investigation of whether the size of fiscal adjustment is systematically related to the composition of the consolidation package, specifically the reliance on current expenditure cuts. Here, for each consolidation episode, the cumulative fiscal adjustment over the episode was plotted on the vertical axis against the ratio of cumulative current expenditure cuts to total fiscal adjustment on the horizontal axis. Figure 1 shows the relationship for CAPB-based episodes, and Figure 2 shows the relationship for narrative-based episodes.

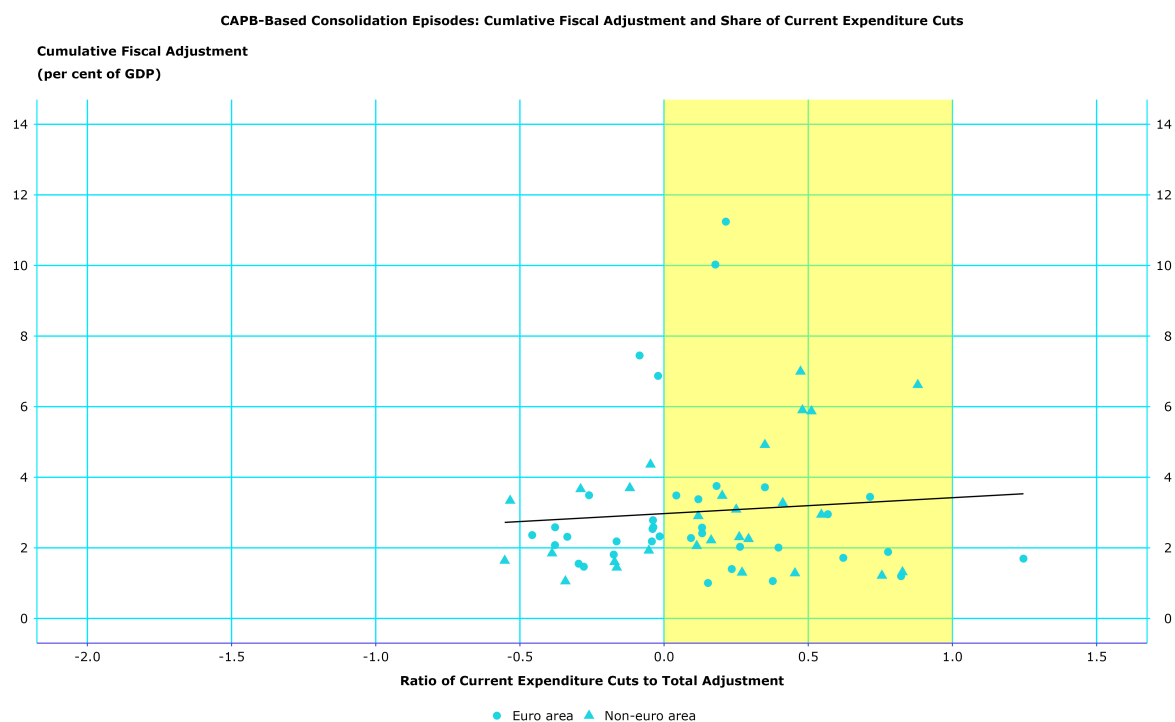


Figure 1: *CAPB-based consolidation episodes: cumulative fiscal adjustment and share of current expenditure cuts*

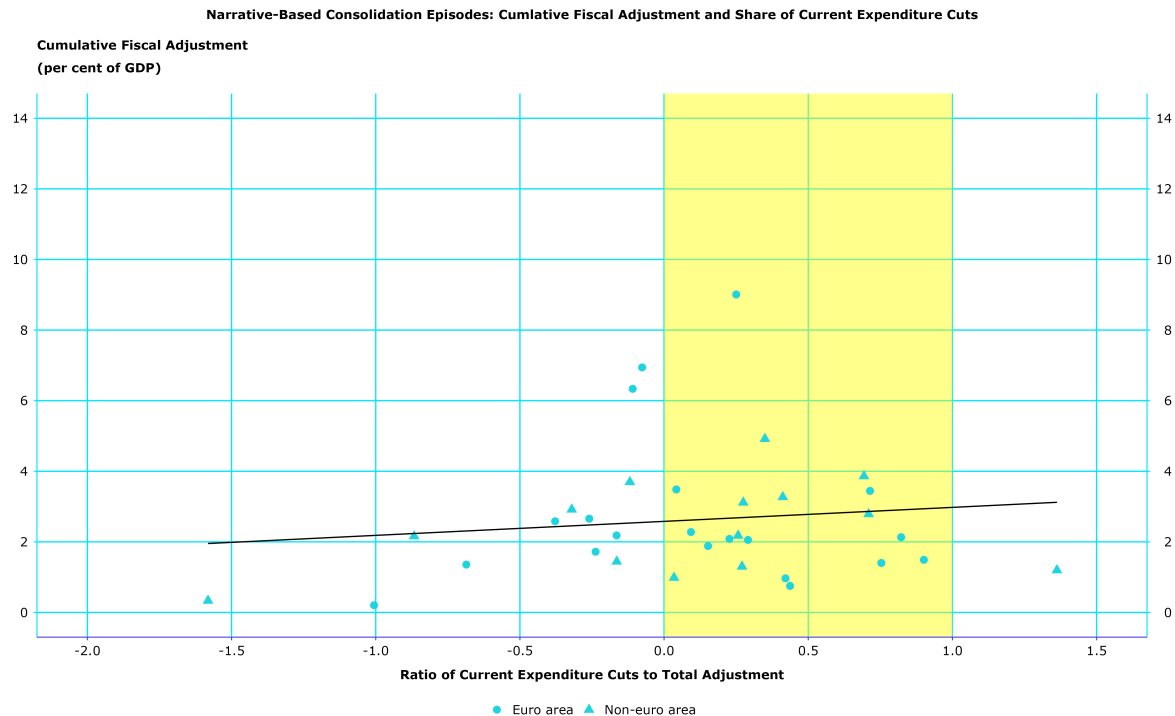


Figure 2: *Narrative consolidation episodes: cumulative fiscal adjustment and share of current expenditure cuts*

Thus, in both the CAPB- and narrative- based episodes, there is a positive relationship between the share of current expenditure cuts and cumulative fiscal adjustment. Despite some heterogeneity, at least descriptively, a higher share of current expenditure cuts is associated with a higher likelihood that consolidation is pursued and with a larger cumulative adjustment. This descriptive result is consistent with Ahrend et al. (2006), who also document a positive relationship between the share of current expenditure cuts and the cumulative size of fiscal adjustment. In their econometric analysis, a higher share of non-capital expenditure cuts increases the likelihood that an adjustment is pursued and is associated with larger cumulative adjustments. The evidence presented here therefore confirms this pattern for CAPB-based episodes and shows that a similar relationship also appears under narrative identification, despite the different timing and composition of the episodes.

Subsequently, by first classifying episodes as expenditure driven when current expenditure cuts account for at least half of the total adjustment, and revenue-driven otherwise, and then comparing the average change in long-term interest-rate spreads and policy-rate indicators during the initial phase and over the whole episode, Figures 3 and 4 show whether monetary and financial conditions respond differently depending on the composition of fiscal consolidation. Figure 3 shows the relationship for CAPB-based episodes, and Figure 4 shows the relationship for narrative-based episodes.

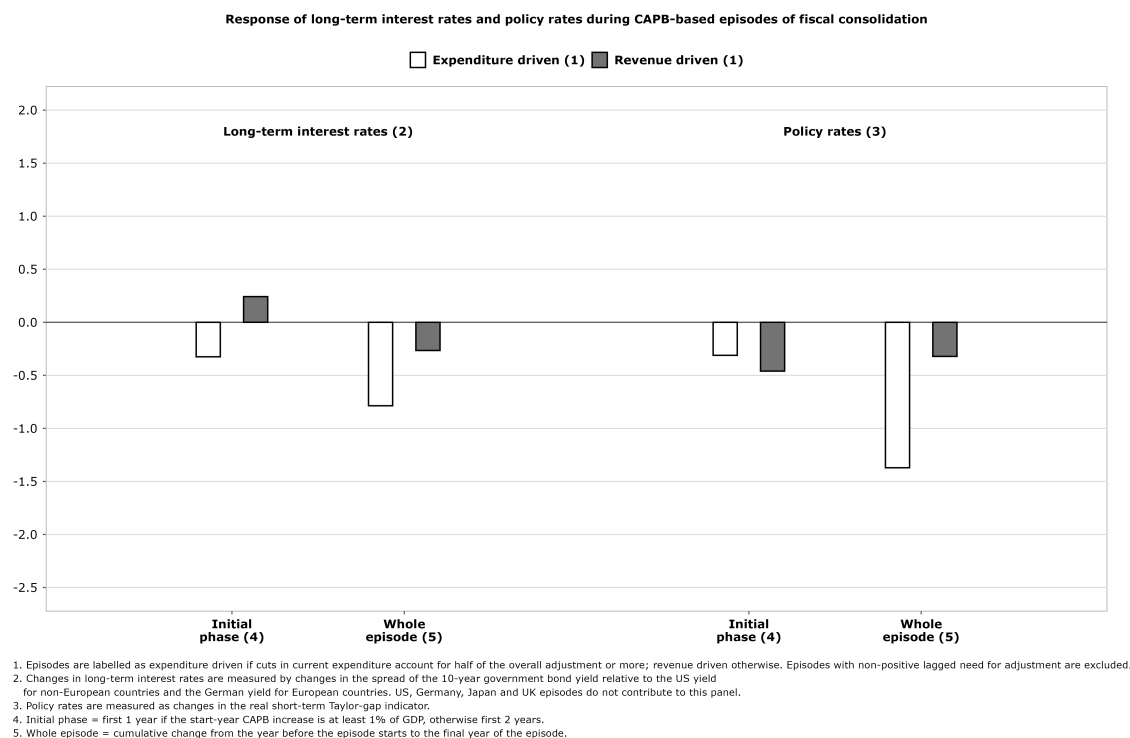


Figure 3: *CAPB-based consolidation episodes: response of long term interest rates and policy rates*

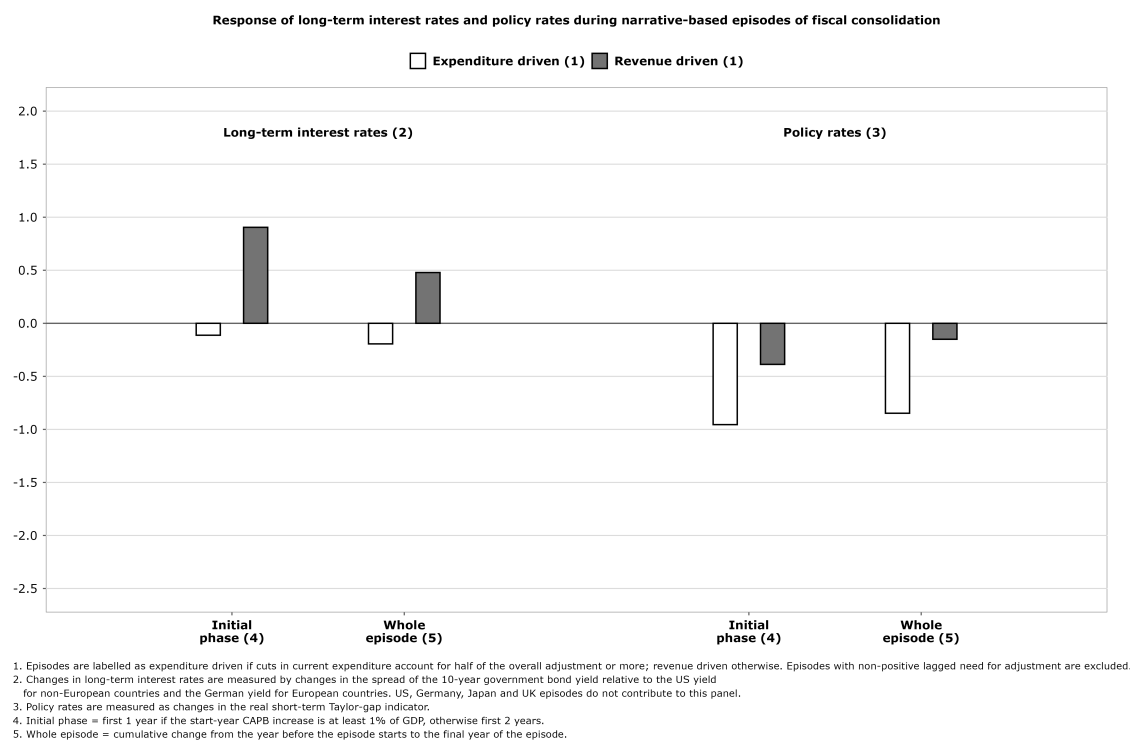


Figure 4: *Narrative consolidation episodes: response of long term interest rates and policy rates*

For both CAPB- and narrative- based consolidations, expenditure-driven episodes are associated descriptively with larger declines in interest rates. Specifically, long-term interest rates fall during both the initial phase and the whole episode for expenditure-driven consolidations, whereas revenue-driven episodes are associated with an initial increase and only a modest decline over the full episode. This pattern is particularly visible for narratively identified exogenous episodes. For policy rates, CAPB-based expenditure-driven episodes show a much stronger decline over the whole episode than revenue-driven episodes. In the narrative sample, both expenditure-driven and revenue-driven episodes are associated with declining policy rates, but the fall is larger for expenditure-driven episodes, especially during the initial phase. These patterns suggest that the composition of fiscal adjustment may affect how financial markets and monetary authorities respond to consolidation. In particular, consolidation packages relying more heavily on expenditure cuts may be perceived as more credible when they contain structural and politically costly measures, such as reforms to pensions, transfers or public-sector wages. This implies that interest-rate movements are not fully exogenous to fiscal consolidation: interest rates may fall because monetary conditions independently support adjustment, but also because credible fiscal packages induce a favourable response from financial markets or monetary authorities. For this reason, the share of current expenditure cuts is included as a proxy for adjustment quality and credibility in the extended specifications, helping to address reverse causality and to better isolate the association between monetary conditions and fiscal consolidation outcomes.

Furthermore, to capture the quality of fiscal consolidation adjustments in a more fine-grained manner, the analysis also distinguishes between current expenditure cuts and capital expenditure cuts. Capital expenditure mainly refers to public investment, such as infrastructure spending, while current expenditure includes recurrent spending items such as public wages, transfers, subsidies, pensions and purchases of goods and services. This distinction follows Ahrend et al. (2006), who find that a higher share of non-capital expenditure cuts is associated with a greater likelihood that adjustment is pursued, while a higher share of capital-expenditure cuts tends to have the opposite effect, although this

result is less statistically robust. The interpretation, however, should remain cautious. Capital-expenditure cuts should not be treated as mechanically less durable or easier to reverse; rather, their relevance lies in the possibility that reductions in public investment may weaken the quality of the adjustment by lowering growth-supporting expenditure. Conversely, some current-expenditure measures, such as pension, transfer or public-sector wage reforms, may require substantial political capital and therefore signal a stronger commitment to fiscal consolidation. For this reason, both current and capital expenditure cuts are included as controls in the multivariate analysis, not as direct measures of durability, but as imperfect proxies for the composition, quality and perceived credibility of the fiscal adjustment.

3.2.3.2 Considerations on the Euro Area

Additionally, given the heterogeneity of the sample, to capture whether euro-area countries behaved systematically differently from the remaining countries across different institutional regimes, several euro-area dummies are implemented. As Ahrend et al. (2006) found that prospective euro-area countries behaved differently before the Maastricht Treaty, during the 1992-1997 qualification period and after entry into the European Monetary Union (EMU), it is of interest to see whether these conclusions also hold for the more extended sample up to 2014 that is used in this analysis.

3.3 Econometric Analysis Set-Up

In order to determine whether specified conditions make it more likely that fiscal consolidation episodes, once commenced, will be continued or successful, and to what extent monetary policy can influence these outcomes, a variety of multivariate regressions are used. Then, the obtained results are compared across CAPB identified consolidation episodes and those narratively identified, to investigate whether there are any inconsistencies across these two methodologies.

Following the econometric set-up of Ahrend et al. (2006) the following regressions

in Table 1 were specified to encompass the various components of the decision-making process associated with austerity measures and the different phases of a given consolidation episode.

Table 1: Regression Description

Relevance	No.	Dependent Variable	Type	Sample
Start	1	Binary variable indicating the start of a consolidation episode	Probit	Pooled sample of yearly observations, excluding those observations where an episode was already in progress
Continuation	2	Binary variable indicating the continuation of an ongoing consolidation episode	Probit	Pooled sample containing yearly observations where an episode was in progress
Success	3	Change in the CAPB	Panel fixed effects	Unpooled sample containing yearly observations where an episode was in progress
	4	Binary variable indicating that an episode was “seriously pursued” after the initial starting period	Probit	Sample including all episodes, with each episode representing one observation
	5	Cumulative change in CAPB over the whole adjustment episode	Truncated	Sample including all episodes, with each episode representing one observation
	6	Deceleration in the pace of debt accumulation resulting from the consolidation episode, measured as the difference between the gross debt-to-GDP ratio in the two years before and after an episode	Robust OLS	Sample including all episodes, with each episode representing one observation
	7	Cumulative length, in years, of an adjustment episode	Truncated	Sample including all episodes, with each episode representing one observation
	8	Cumulative change in the CAPB in the two years after a consolidation episode	Robust OLS	Sample including all episodes, with each episode representing one observation

4 Data

4.1 Identified Episodes of Fiscal Consolidation

First, it is of interest to compare the composition of the fiscal consolidation episodes that are identified using the two different methods.

Table 2: Episodes of Fiscal Adjustment (Narrative Identification)

Country (1978–2014)	Period	No. episodes	No. years
Australia	85, 93–94, 95–99	3	8
Austria	80, 84, 96–97, 01, 11–13, 14	6	9
Belgium	82, 86–87, 90, 92–95, 96–98, 10	6	12
Canada	83, 85–90, 10–14	3	12
Denmark	82, 94, 09, 12–14	4	6
Finland	92, 10–11, 12–13, 14	4	6
France	79, 87, 91, 95–96, 10–12	5	8
Germany	82, 91–92, 93–94, 97–99, 03–04, 06–07, 11–14	7	16
Ireland	82, 08	2	2
Italy	91–93, 04, 09	3	5
Japan	79, 97, 03–06	3	6
Portugal	83, 00, 02, 05–07, 10–13	5	10
Spain	83, 89, 92, 09–14	4	9
Sweden	84, 93–98	2	7
United Kingdom	79–80, 81–82, 93–95, 96–99, 10–11	5	13
United States	78, 81, 85, 87–89, 90, 93–98, 11–12, 13–14	8	17
<i>16 countries</i>		<i>70</i>	<i>146</i>

Source: IGIER Database on Fiscal Adjustment Plans (IGIER, 2026)

As shown in Table 2, through the narrative approach, over the period between 1978 and 2014, across the 16 countries, a total of 70 exogenous austerity episodes were identified. The number of episodes by country ranged from 2 to 8, with an average of approximately 4. The collective number of years over which countries introduced and implemented austerity measures ranged from 5 to 17 years, with an average of approximately 9 years, in the time period considered.

Table 3: Episodes of Fiscal Adjustment (CAPB Identification)

Country (1978–2014)	Period	No. episodes	No. years
Australia	95–99, 02–03, 05–06, 13–14	4	11
Austria	92, 96–97, 01, 05, 10–13	5	9
Belgium	87, 89–90, 93–97, 98, 06, 12–13	6	12
Canada	79–80, 81, 86–90, 94–97, 11–14	5	16
Denmark	86–88, 96–99, 04–05, 13–14	4	11
Finland	88, 94, 96–97, 98, 00	5	6
France	96, 11–12	2	3
Germany	92–94, 96–99, 07, 11–14	4	12
Ireland	96–98, 03–04, 11–14	3	9
Italy	90–93, 95–97, 07, 12–13	4	10
Japan	97, 99, 04–05, 06, 14	5	6
Portugal	92, 95, 02, 06–07, 11–13	5	8
Spain	86–87, 92, 96–97, 10–14	4	10
Sweden	86–87, 96–98, 00, 04–05	4	8
United Kingdom	80–81, 82, 94–99, 10–11, 13	5	12
United States	87–89, 93–98, 05–06, 11–14	4	15
<i>16 countries</i>		69	158

Source: OECD Economic Outlook Database (OECD, 2026)

Instead, as shown in Table 3, through the CAPB-based approach, over the same sample, a total of 69 cyclically-adjusted discretionary austerity episodes were identified. The number of episodes by country ranged from 2 to 6, with again an average of approximately 4. The collective number of years over which countries introduced and implemented austerity measures ranged from 3 to 16 years, with a slightly higher average of approximately 10 years, across the time period considered.

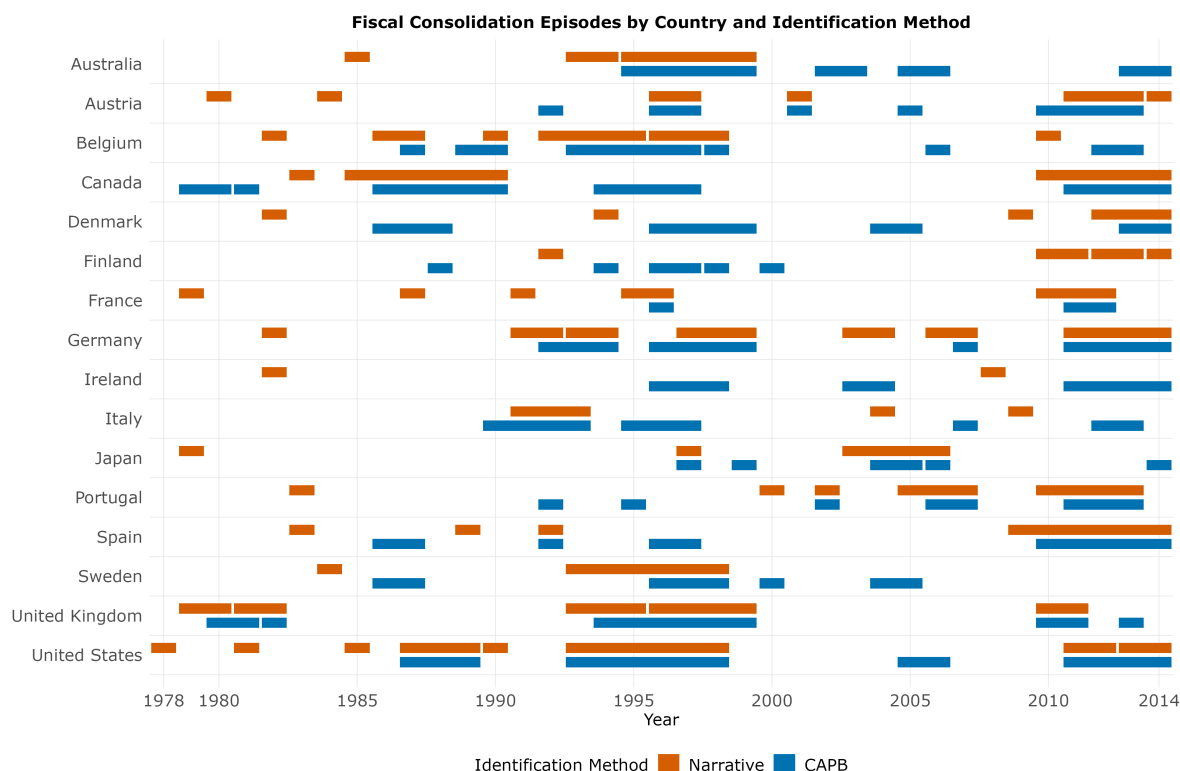


Figure 5: *Identified fiscal consolidation episodes*

Structural differences instead are highlighted in Figure 5 above. Overall, whilst the two identification methods often detect the same broad consolidation waves, they don't consistently match on exact years, duration and country coverage. The narrative- and CAPB- based approaches agree most clearly in the mid 1990s, suggesting robustness across these two methods in what can be considered as an institutionally meaningful time period, especially for Europe, with regards to the Maastricht criteria prior to the EMU. The two are also relatively aligned in the post-2010 period, identifying many consolidation episodes after the global financial crisis and during the European sovereign debt crisis.

In general, however, episodes identified using the CAPB are often more continuous whereas narrative episodes are often shorter and more fragmented. This is in part expected by construction, as policy-driven consolidation years can show several separate episodes despite the fiscal balance changing more smoothly over time. Further, it is interesting to note that some countries show strong disagreement between methods. For example in Ireland, the narrative method identifies only two brief consolidation episodes, whereas the CAPB approach suggests much longer consolidation periods, especially around the

late 1990s and after 2010. Other similar mismatches can be seen in Japan, Italy and Canada, amongst others, demonstrating that the two methods are capturing related but distinct events. Thus, this graph clearly highlights the discrepancies that result from outcome-based identification and policy-action-based identification.

Ultimately, CAPB and narrative methods do not produce identical treatment periods, motivating potentially different conclusions that may be established regarding the effects of monetary policy on fiscal consolidation. Specifically, the CAPB-based identification may capture changes driven by macroeconomic conditions, forecasting errors, revenue windfalls, or automatic stabilisers that are not fully removed by cyclically adjusting the budget balance. Narrative identification instead, is closer to policy intent, but it may miss gradual fiscal improvements not tied to clearly documented policy announcements.

4.2 Econometric Analysis Results

Table 4 summarises the main comparative results of the multivariate analysis. It reports, for each dependent variable, the direction and statistical significance of the estimated effects, allowing for a direct comparison between CAPB-based and narrative-based consolidation episodes. The results are then discussed in detail in the following section.

Table 4: Summary Results of Multivariate Analysis

		Indicator of "Need for Adjustment"	Real Exchange Rate	Share of Expenditure Cuts	Share of Capital Expenditure Cuts	Output Gap	Change in Nominal Short Rate	Change in Real Short Rate	Change in Real Short Rate Relative to Taylor Rule	Change in Nominal Long Rate	Change in Real Long Rate
Probability that episode is started	CAPB	+++							--		
	Narrative	+	---				--	--			
Probability that episode is continued (yearly basis)	CAPB	++				-					
	Narrative	+									
Size of adjustment (yearly basis)	CAPB	+++		+		-				+	+
	Narrative	+++					+	+		++	++
Probability that episode is seriously pursued (after initial period)	CAPB	++		+++		---				-	
	Narrative	-		++		--					
Size of cumulative adjustment (total episode)	CAPB	+++		++	++	-					
	Narrative					-	-		+++		
Deceleration in pace of debt accumulation (total episode)	CAPB		-			---	--	--			
	Narrative	+	--		--	-					
Length of episode	CAPB										
	Narrative		+						+		
Extent of slippage in aftermath of episode	CAPB			-		++			++		
	Narrative	-			++	+					

Note: The signs indicate the direction of the estimated effect of each explanatory variable on the dependent variable. Only statistically significant coefficients are reported. One, two and three signs denote significance at the 10%, 5% and 1% levels, respectively.

Overall, the two identification methods produce different results, reflecting the fact that the CAPB- and narrative-based approaches do not identify identical treatment periods. Although the total number of episodes is similar across the two methods, the underlying composition of the samples differs in timing, duration and country coverage. This is therefore reflected in the estimated determinants of the start, continuation and success of fiscal consolidation episodes.

With regards to the start of consolidation episodes, the strongest result under CAPB-based identification is the positive and highly significant association between the “need for adjustment” and the probability that an episode begins. Under narrative-based identification, the same variable is also positive, but the result is weaker. The output gap does not appear to play a meaningful role under either method, whereas the real exchange rate is negative and statistically significant only in the narrative specifications. Monetary variables matter only in a limited way. In the CAPB-based regressions, the only significant monetary indicator is the change in short rates relative to the Taylor rule, which enters negatively. By contrast, in the narrative-based regressions, only nominal and real short-rate changes are significant. Thus, easier monetary conditions appear to be associated with the start of consolidation episodes, but the relevant monetary channel differs across the two identification methods.

With regards to continuation, the evidence is much narrower. The only variable that is significant under both identification methods is the indicator of “need for adjustment”, which enters positively for both CAPB- and narrative- based episodes, although the association is stronger under the former. The output gap is significant only under CAPB-based identification, where it enters with a negative sign. Most importantly, no monetary variable is significant for continuation under either identification method. Therefore, the continuation of austerity measures appears to be explained mainly by fiscal pressure, and in the CAPB case also by weak cyclical conditions, rather than by monetary conditions.

Finally, the regressions addressing the success of fiscal consolidation episodes do not point to a single uniform pattern. Under CAPB-based identification, the “need for ad-

justment” is repeatedly significant, especially for yearly and cumulative adjustment, while the output gap and the composition of adjustment also matter in several specifications. The share of expenditure cuts is positively associated with yearly adjustment, serious pursuit and cumulative adjustment, and negatively associated with post-episode slippage. Monetary variables, however, are significant only in selected cases. Falling nominal and real short rates are associated with a stronger deceleration in debt accumulation, while the Taylor-rule measure is significant for post-episode slippage. Beyond these specific results, there is no stable evidence that easier monetary conditions increase the size, duration or durability of CAPB-based consolidations.

Under narrative-based identification, the evidence for a supportive monetary channel is much weaker and more ambiguous. Monetary variables are generally insignificant for continuation, serious pursuit, debt deceleration and post-episode slippage. In some success specifications, the signs are also difficult to interpret as monetary accommodation: long-rate increases are associated with larger yearly adjustment, while the Taylor-rule measure enters positively for cumulative adjustment and episode length. Therefore, although short-rate easing is associated with the start of exogenous consolidation episodes, the broader narrative-based results do not suggest that easier monetary conditions systematically improve consolidation outcomes.

Overall, the results provide almost no support for the view that monetary easing improves fiscal consolidation. Fiscal need, weak cyclical conditions, the real exchange rate, euro-area institutional regimes and the composition of adjustment appear more frequently as significant determinants. Monetary variables matter in some specifications, especially for the start of episodes and selected debt-dynamic outcomes, but their effect is not consistent across the start, continuation and success of consolidation, nor across CAPB- and narrative-based identification.

The full regression outputs are reported in the appendix, while the following section discusses the implications of these results in detail.

5 Discussion

5.1 The Start of Consolidation Episodes

The first stage of the multivariate analysis considers whether the macroeconomic and monetary environment affects the probability that a fiscal consolidation episode begins. Overall, the start of consolidation appears to be influenced by monetary conditions under both identification methods, but the relevant monetary channel differs depending on whether episodes are identified using the CAPB or the narrative approach.

Under the CAPB-based classification, the most robust result is the positive and highly significant coefficient on the lagged “need for adjustment”, which remains approximately equal to 0.02 in average marginal effect terms across all specifications. This indicates that CAPB-based consolidation episodes are more likely to start when the preceding fiscal position is weaker and the pressure for adjustment is stronger. By contrast, the lagged output gap and the change in the real exchange rate are not statistically significant, suggesting that the start of a CAPB-identified episode is not primarily explained by cyclical conditions or exchange-rate movements once fiscal need is controlled for. Further, the euro-area dummy for 1992-1997 is positive and highly significant in every specification, implying that prospective euro-area countries were much more likely to begin consolidation during the Maastricht qualification period.

The role of monetary variables under CAPB identification is limited but relevant. Neither changes in nominal or real short rates, nor changes in nominal or real long rates are significant. The only significant monetary indicator is the change in short rates relative to the Taylor rule, which has a negative relationship: a tightening relative to the Taylor-rule benchmark reduces the probability of starting an episode, while easier-than-rule monetary conditions increase it. Thus, for CAPB-based episodes, what matters is not simply whether interest rates are falling, but whether monetary policy is easier than would be implied by the cyclical position of the economy. These results are therefore only partly aligned with Ahrend et al. (2006): while they also find that standard interest-rate

changes do not explain the start of consolidation episodes, they do not find a significant role for the Taylor-rule-based monetary indicator either.

Under narrative-based identification, the results differ meaningfully. The “need for adjustment” variable is still positive, as expected, but its significance is weaker: it becomes statistically significant only in specifications that include the real exchange rate and individual interest-rate variables, and loses significance once the output gap or the Taylor-rule measure is included. This suggests that exogenously identified consolidation episodes are only partly explained by the underlying fiscal pressure captured by the synthetic “need for adjustment” indicator. The output gap is not statistically significant, indicating that the start of an exogenous consolidation does not appear to be strongly associated with the cyclical position of the economy. By contrast, the lagged change in the real exchange rate is negative and highly significant across all specifications, which implies that exogenous consolidation episodes are more likely to start after a real depreciation. This is consistent with the idea that narrative fiscal consolidations may often be introduced in response to external pressure or confidence concerns, rather than simply in response to weak public finances. Unlike in the CAPB-based results, the euro-area dummies are not statistically significant, meaning that the Maastricht qualification period does not appear to have made euro-area countries significantly more likely to intentionally start consolidation episodes.

With regards to monetary variables, the strongest narrative-based result concerns short-term rates: both lagged nominal and real short-rate changes enter negatively and significantly. This implies that decreases in short-term interest rates are associated with a higher probability that an exogenous consolidation episode starts. However, this result is not robust across the full set of monetary proxies, since changes in nominal and real long rates are insignificant, and the change in short rates relative to the Taylor rule is also insignificant. Therefore, for narratively identified episodes, the evidence suggests that easier short-term monetary conditions may accompany the beginning of consolidation, but this cannot be generalized to broader monetary accommodation once long rates or Taylor-rule deviations are considered.

Overall, both identification methods provide some evidence that easier monetary conditions are associated with the start of consolidation, but they point to different channels. In the CAPB case, the relevant channel is relative monetary accommodation, measured through short rates relative to a Taylor-rule benchmark. In the narrative case, the relevant channel is direct short-rate easing. However, long rates are insignificant under both identification methods, suggesting that the beginning of consolidation is not strongly driven by financial-market rates or broad improvements in market confidence. Thus, the evidence for the start phase supports a limited, rather than strong, monetary policy interaction.

5.2 The Continuation of Consolidation Episodes

The second stage of the analysis considers whether an already-started consolidation episode continues from the previous year. Here, the evidence is much clearer: monetary variables do not significantly explain continuation under either identification method. Instead, continuation appears to depend mainly on remaining fiscal pressure, with some additional evidence for weak cyclical conditions under CAPB identification and euro-area institutional effects under narrative identification.

Under the CAPB-based classification, fiscal need remains relevant. The lagged “need for adjustment” is positive in all specifications and is mostly statistically significant, with average marginal effect coefficients around 0.03. This suggests that governments are more likely to continue consolidating when the underlying fiscal imbalance remains large. Further, the output gap enters negatively and is significant in almost all specifications, indicating that continuation is more likely when economic conditions are weak, or when output is below potential. This is consistent with the idea that fiscal pressure and crisis conditions can strengthen the political incentive to maintain consolidation.

However, the real exchange rate, the quality of adjustment and all monetary indicators are insignificant. Additionally, the euro-area dummy for the period after 1998 is negative and often weakly significant, suggesting that, after EMU entry, euro-area coun-

tries were less likely to continue consolidation than the rest of the sample, at least when considering CAPB-identified episodes. Compared with the conclusions reached by Ahrend et al. (2006), this is an important divergence, as they instead find that falling short-term rates and easier monetary conditions relative to a Taylor rule increase the likelihood that consolidation is continued. This could be due to the fact that the episodes identified in the sample differ, as not all 25 countries are included and the time period considered in this analysis extends up to 2014 (instead of 2005).

Under narrative-based identification, the continuation results are also narrow and mostly dependent on fiscal and institutional variables, rather than monetary ones. The lagged “need for adjustment” is positive and statistically significant in almost all specifications, with an average marginal effect of around 0.02. This implies that, once a narrative consolidation episode has begun, it is more likely to continue when the fiscal position remains under pressure. However, the output gap is negative but not statistically significant, suggesting that the continuation of narratively identified episodes is not systematically driven by whether the economy is above or below potential. The real exchange rate relative to its historical average is also insignificant, so there is no evidence that exchange-rate valuation affects whether these episodes continue.

The most notable non-fiscal result concerns the euro-area dummies. Both the 1992-1997 and post-1998 euro-area dummies are negative and frequently statistically significant. This suggests that, conditional on an exogenous consolidation having started, euro-area countries were less likely to continue such episodes during the Maastricht qualification period and after EMU entry. This is different from what might have been expected for the 1992-1997 period, and may reflect the fragmented nature of narrative episodes, where new policy plans are classified as new starts rather than continuations.

Most importantly, none of the monetary variables are statistically significant. Nominal and real short-rate and long-rate changes, and the Taylor-rule measure all fail to explain continuation. Adding the share of expenditure cuts and the share of capital-expenditure cuts does not change this conclusion, as quality-of-adjustment proxies are also insignificant. Therefore, the continuation of narrative-based consolidation episodes

appears to be explained mainly by remaining fiscal pressure and euro-area institutional-period effects, not by monetary conditions.

Overall, the continuation results provide no support for a systematic monetary accommodation channel. Monetary easing may help create the conditions under which consolidation begins, but maintaining adjustment year after year appears to depend more on persistent fiscal pressure, institutional constraints and credibility concerns. This is one of the clearest differences between the present results and the conclusions reached by Ahrend et al. (2006).

5.3 The Success of Consolidation Episodes

The final stage of the analysis considers whether monetary conditions affect the success of fiscal consolidation episodes. This is measured through several outcomes: the size of yearly adjustment, the probability that an episode is seriously pursued, the size of cumulative adjustment, the deceleration in the pace of debt accumulation, the length of the episode and the extent of post-episode slippage. Overall, the success results do not point to a single uniform pattern. Monetary variables matter only in selected specifications, while fiscal need, weak cyclical conditions and the composition of adjustment appear more frequently as significant determinants.

With regards to the size of yearly fiscal adjustment, the dominant result under CAPB-based identification is again the “need for adjustment”, which is positive and highly significant across all specifications. This means that larger annual improvements in the CAPB occur when the initial fiscal imbalance is greater. The output gap is negative but only becomes statistically significant in some specifications, particularly when expenditure-composition controls are added. This provides some evidence that yearly adjustment is larger under weaker cyclical conditions, although the result is less robust than for fiscal need. Importantly, the composition of adjustment also matters: the share of expenditure cuts is positive and weakly significant, indicating that expenditure-based consolidations are associated with larger annual CAPB improvements. This reinforces the idea that the

structure of the fiscal package is relevant, not only the size of the fiscal imbalance. The monetary results, by contrast, are more surprising and do not point to a straightforward role for monetary accommodation. Changes in nominal and real short rates are positive but insignificant, while changes in nominal and real long rates are positive and weakly significant. Therefore, larger yearly adjustments are associated with increases, rather than decreases, in long-term rates. This finding is difficult to reconcile with the view that easier monetary conditions support consolidation, and is instead more consistent with a financial-market pressure effect, whereby higher long-term rates increase the urgency of fiscal adjustment.

Under narrative-based identification, yearly adjustment is also most clearly explained by fiscal need. The lagged “need for adjustment” is positive and statistically significant across all specifications. The output gap is negative, but its significance depends on the specification, suggesting that weaker cyclical conditions are sometimes associated with a larger yearly adjustment. Unlike in the CAPB-based results, the share of current expenditure cuts and capital-expenditure cuts are not statistically significant, implying that within the narrative sample, the annual size of observed CAPB adjustment is not strongly explained by whether the fiscal package is more expenditure-based. The monetary variables again do not support a simple accommodation story. Lagged nominal and real short-rate changes are positive and weakly significant when included alone, but lose significance once composition controls are added. By contrast, nominal and real long-rate changes are positive and statistically significant even after composition controls are included. This is a surprising result: if larger narrative-based consolidations were immediately perceived as credible and risk-reducing, one would expect them to be associated with falling, not rising, long-term rates. Since long-term interest rates are not a pure measure of sovereign risk premia, this result should be interpreted cautiously. However, it provides no evidence in this specification that consolidation reduces the risk premium; instead, it is more consistent with the view that stronger yearly adjustments occur when financial-market pressure is already high.

When considering the probability that an episode is seriously pursued* after the initial phase, the CAPB-based results show that the output gap is the most consistently significant macroeconomic variable. The output gap enters negatively and is highly significant across specifications, showing that episodes are more likely to be seriously pursued when the economy is initially weaker. The “need for adjustment” is positive throughout, but is only statistically significant once quality-of-adjustment controls are included, suggesting that fiscal pressure matters more when the credibility of the consolidation package is accounted for. Most importantly, the composition of the adjustment package plays a central role: the share of expenditure cuts is positive and highly significant across all specifications. This indicates that CAPB-based episodes relying more heavily on expenditure cuts are systematically more likely to be pursued beyond the initial phase. By contrast, monetary variables provide weaker results. Average changes in nominal and real short rates during the starting phase are positive but insignificant, while nominal and real long-rate changes are negative but insignificant. The Taylor-rule measure is negative and weakly significant under a full specification, supporting limited evidence that easier-than-rule monetary conditions during the starting phase increase the probability that an episode is seriously pursued. Compared with the results obtained by Ahrend et al. (2006), these findings are broadly consistent, despite weaker statistical strength.

For narrative-based episodes, the serious-pursuit specification requires particular caution because the sample is extremely small, with only 13 observations. Within this limitation, the output gap again emerges as the most important macroeconomic variable: it enters negatively and significantly in many specifications, implying that episodes are more likely to be seriously pursued when the economy is weaker at the end of the starting phase. This result is somewhat surprising, since weaker economic conditions might normally be expected to make governments less willing to continue consolidation. A plausible interpretation is that, in these cases, weak cyclical conditions capture a broader crisis environment in which fiscal adjustment becomes more urgent and harder to abandon, rather

*An episode is classified as “seriously pursued” if, in the two years following its start, an additional adjustment of at least one percentage point of GDP is achieved

than an environment in which consolidation is economically easier to pursue. The “need for adjustment” enters negatively and is significant in several specifications, which is also counterintuitive because it implies that stronger remaining fiscal pressure is associated with a lower probability of serious pursuit. However, this should not be interpreted too strongly given the very small sample and the instability of the estimates. By contrast, the composition result is more consistent with the broader evidence on adjustment quality: the share of expenditure cuts is positive and significant in the specification where it can be meaningfully estimated, indicating that expenditure-based narrative episodes are more likely to be seriously pursued. This suggests that, even under narrative identification, the structure of the consolidation package matters for whether an episode continues beyond its initial phase. However, the significance disappears in later columns where coefficients collapse to zero, again reflecting the limited variation in the sample. Similarly, with regard to monetary variables, there is no significant evidence that interest-rate movements affect the probability of serious pursuit.

The size of cumulative adjustment over the whole episode also differs across identification methods. Under CAPB-based identification, the strongest and most stable result is the “need for adjustment”, which is positive and highly significant across all specifications. In Table A.5.2, the coefficient ranges from 0.78 to 0.92, implying that a one-percentage-point increase in the initial “need for adjustment” is associated with roughly a 0.8 to 0.9 percentage-point larger cumulative CAPB improvement over the episode. This should be interpreted as a continuous association, rather than as evidence of a precise debt or primary-balance threshold beyond which consolidation size increases discontinuously. The output gap also enters negatively and is statistically significant in several specifications. In the baseline specification, for example, the coefficient is -0.98, implying that a one-percentage-point lower output gap is associated with almost a one-percentage-point larger cumulative adjustment. This result is somewhat surprising, since weak economic conditions would normally be expected to make large fiscal consolidations harder to sustain. A plausible interpretation is that a weak cyclical position captures a broader crisis environment in which the pressure to consolidate becomes stronger, rather than an envi-

ronment in which adjustment is economically easier. The composition of the adjustment episode is also highly relevant: the share of expenditure cuts is positive and significant across the specifications in which it is included, with coefficients between approximately 4.5 and 5.7. Monetary variables, by contrast, are not statistically significant, despite generally entering with negative signs consistent with a possible role for monetary easing in supporting larger adjustment. This means that, at least under CAPB-based identification, larger cumulative adjustment is more closely associated with fiscal pressure, weak cyclical conditions and the composition of fiscal action than with monetary conditions. Compared with Ahrend et al. (2006), this sample therefore fails to replicate the main monetary conclusion that falling interest rates are associated with larger cumulative adjustments.

Under narrative-based identification, the “need for adjustment” is not statistically significant in any cumulative-adjustment specification, despite being positive in most columns. This suggests that stronger initial fiscal pressure does not reliably explain the total amount of consolidation ultimately achieved in narratively identified episodes. The output gap is also mostly insignificant, but becomes negative and highly significant in the specifications that include the Taylor-rule monetary variable. Composition variables do not explain cumulative adjustment, as neither the share of current expenditure cuts nor the share of capital-expenditure cuts is significant. The monetary variables are mixed. The average change in the nominal short rate during the episode is negative and becomes weakly significant once composition controls are included, suggesting that falling nominal short rates may be associated with larger cumulative adjustment. However, the real short-rate measure and long-rate changes are not significant. More importantly, the cumulative change in the short rate relative to the Taylor rule is positive and highly significant in both specifications. This means that larger cumulative adjustment is associated with a tightening relative to the Taylor-rule benchmark, not with easier-than-rule monetary conditions. Therefore, the narrative evidence for cumulative adjustment is ambiguous and vividly prevents a clean conclusion that monetary accommodation improves the cumulative success of exogenously identified episodes.

With regards to the deceleration in the pace of debt accumulation, the CAPB-based

results provide the clearest evidence of a monetary channel. The “need for adjustment” is not significant, indicating that the initial fiscal imbalance cannot directly explain whether a CAPB-identified episode improves debt dynamics. However, the output gap is negative and highly significant, indicating that episodes beginning in weaker cyclical conditions are followed by a stronger deceleration in debt accumulation. The real exchange rate is also negative and weakly significant, suggesting that a weaker exchange-rate position may be associated with stronger improvements in debt dynamics, possibly through support from export demand. The euro-area dummy for 1992-1997 is positive and significant, consistent with strong debt-dynamics improvement during the Maastricht qualification period. More importantly, nominal and real short-rate changes are negative and significant, both when considered alone and when composition controls are added. This means that falling short-term rates during the identified episode are associated with a larger slowdown in debt accumulation. Therefore, easier monetary conditions matter, but only through short-term interest rates, and only when considering the evolution of debt dynamics as a measure of success.

Under narrative-based identification, the debt-dynamics results are weaker. The “need for adjustment” is positive and weakly significant in only a few specifications, while the output gap is negative and significant in several specifications. The real exchange rate is also negative and significant, indicating that a more depreciated real exchange-rate position is associated with a stronger improvement in debt dynamics. The share of capital-expenditure cuts is negative and statistically significant, suggesting that narrative episodes relying more heavily on cuts to capital expenditure are associated with a weaker deceleration in debt accumulation. This should not be interpreted as evidence that capital-expenditure cuts are mechanically less durable or easier to reverse. Rather, a more cautious interpretation is that reductions in public investment may weaken the quality of adjustment by lowering growth-supporting expenditure. With regard to monetary variables, there is no statistically significant evidence that changes in interest rates improve debt dynamics. Short-rate and long-rate changes generally enter with negative signs, which would be consistent with falling rates supporting debt deceleration, but none

of these coefficients is significant. Therefore, unlike the CAPB-based results, the narrative-based debt-dynamics regression does not provide evidence that easier monetary conditions significantly improve the debt outcome of consolidation episodes. This divergence is not merely a difference in statistical significance, but reflects the broader identification problem at the centre of this paper; the next section therefore examines why CAPB-based and narrative-based episodes can produce different estimates of the monetary-fiscal relationship.

The evidence for episode length is also weak. Across all CAPB-based specifications, there are no statistically significant results. More importantly, none of the monetary variables are significant, meaning that none of the relationships are precise enough to support a conclusion about the role of monetary conditions in determining or influencing episode length. This differs from Ahrend et al. (2006), who are able to conclude that longer episodes are associated with fiscal need, expenditure-based adjustments during adverse cyclical conditions and falling interest rates. Under narrative-based identification, the “need for adjustment” and output gap are also insignificant throughout. The real exchange rate becomes only weakly significant in the final specification, and the post-1998 euro-area dummy is negative and weakly significant in the final specification. The only monetary variable that becomes significant is the cumulative change in the short rate relative to the Taylor rule, which enters positively and weakly significantly once composition controls are included. This means that longer narrative episodes are associated with tighter-than-rule monetary conditions, not easier ones. Therefore, the duration results do not support the hypothesis that monetary easing lengthens consolidation efforts.

Finally, the post-episode slippage results again suggest that durability is more closely related to cyclical conditions and adjustment composition than to monetary conditions. Under CAPB-based identification, there is no evidence that countries with strong remaining fiscal pressure experience less slippage. Instead, the output gap is positive and statistically significant in most specifications, implying that countries ending consolidation in better cyclical conditions are more likely to loosen afterward. This is plausible, as stronger economic conditions may reduce the perceived urgency of maintaining tight

fiscal policy, thereby creating room for fiscal relaxation. The share of expenditure cuts is negative and weakly significant, suggesting that expenditure-based consolidations are less prone to post-episode slippage. The monetary variables show a specific but limited result: cumulated changes in nominal and real, short and long, rates are all insignificant, but the cumulated change in short rates relative to the Taylor rule is positive and significant. This indicates that tighter-than-rule monetary conditions during the episode are associated with greater subsequent slippage.

Under narrative-based identification, the “need for adjustment” is generally insignificant, although it becomes negative and weakly significant in one specification. The output gap is positive and weakly significant in some specifications, again indicating that countries ending consolidation in stronger cyclical conditions may be more likely to loosen afterward. The cumulative size of the adjustment is not significant, and the broad share of expenditure cuts is also insignificant. However, the share of capital-expenditure cuts is positive and significant, meaning that episodes relying more heavily on capital-expenditure cuts are followed by greater post-episode deterioration in the CAPB. This should not be interpreted as evidence that capital-expenditure cuts are necessarily less durable or less credible. Rather, it suggests that the composition of expenditure cuts matters for the durability of adjustment, and that packages relying heavily on reductions in public investment may be associated with weaker post-episode fiscal performance. Most importantly, none of the monetary variables are significant. Therefore, the narrative-based slippage results suggest that post-episode durability is more closely related to cyclical conditions and the composition of adjustment than to monetary conditions.

Overall, the success regressions provide only very limited support for the view that easier monetary conditions improve fiscal consolidation outcomes. Under CAPB identification, there is some evidence that falling short-term rates improve debt dynamics and that tighter-than-rule monetary conditions increase post-episode slippage, but monetary variables are not consistently significant across the full set of success measures. Under narrative identification, the evidence is even weaker and more ambiguous. Monetary variables are insignificant for serious pursuit, debt deceleration and slippage, while long-rate

increases are associated with larger yearly adjustment and Taylor-rule-relative tightening is associated with larger cumulative adjustment and longer episode duration. Therefore, monetary conditions do not appear to be a stable determinant of consolidation success. Fiscal need, weak cyclical conditions, exchange-rate position and the composition of adjustment, instead, provide more consistent explanations.

5.4 The Importance of Identification Methods

The comparison between the CAPB-based and narrative-based results shows that the estimated role of monetary policy depends strongly on how fiscal consolidation is identified. Essentially, the lack of consistency points towards an overall minor, perhaps even negligible, role of accommodative monetary policy in fiscal consolidation. Across the start, continuation and success of fiscal adjustment episodes, monetary variables do not exert a uniform nor consistently supportive effect. Instead, their role is intermittent, specification-dependent and sensitive to the identification method used. This implies that monetary conditions affect fiscal consolidation in some cases, but not in the broad and systematic way suggested by Ahrend et al. (2006). The results are therefore closer to a conditional interpretation: monetary easing may matter at specific moments and through selected channels, especially short-term rates, but the evidence for these 16 countries between 1978 and 2014 does not support the conclusion that easier monetary conditions generally make consolidations more likely to be sustained and successful.

A first explanation concerns the difference between outcome-based and policy-action-based identification. The CAPB approach identifies episodes through observed improvements in the fiscal balance. These improvements are more sensitive to the broader macroeconomic environment, because monetary easing can support output and revenues, reduce the interest burden, and therefore help generate the CAPB improvements that define or sustain the episode. This may explain why the CAPB-based results provide somewhat more support for a favourable monetary-policy role, especially through short rates and Taylor-rule deviations in selected specifications. However, this also means that

the measured monetary effect may partly capture endogenous improvements in fiscal outcomes rather than an independent monetary influence on exogenous fiscal policy.

By contrast, the narrative method identifies consolidation based on policy intent. Since these episodes are not defined by realized fiscal-balance improvements, they may be less mechanically linked to the macroeconomic effects of monetary policy. A narratively identified fiscal plan can be introduced during easier monetary conditions, but whether it produces a large observed CAPB improvement depends on implementation, growth, revenues and the composition of the measures. This may explain why monetary variables are less consistently significant in the narrative success regressions. Under narrative identification, short-rate easing is associated with the start of exogenous episodes, but it does not systematically explain whether those episodes continue or succeed.

A second explanation concerns reverse causality and financial-market pressure. In several specifications, especially those involving yearly adjustment size, higher long-term rates are associated with larger fiscal adjustment. This appears under both identification methods, but is stronger in the narrative results. Such a finding is difficult to interpret as evidence that easier monetary conditions facilitate consolidation, since the sign points in the opposite direction. Instead, it suggests that long-term interest rates may partly capture financial-market pressure on governments to consolidate. Rising long-term rates may indicate market concern about fiscal sustainability, thereby increasing the urgency of fiscal correction and inducing larger adjustments. Conversely, falling long-term rates should not necessarily be interpreted as an independent monetary condition that causes consolidation to succeed. They may instead reflect improved market confidence after a credible consolidation package has already been announced or implemented. This reverse-causality channel is central to the interpretation of the results, because it implies that the observed relationship between interest rates and fiscal adjustment may partly run from fiscal credibility to interest rates, rather than from monetary accommodation to fiscal success.

This is why the composition of adjustment is important. Expenditure-driven consolidations are associated with larger declines in long-term interest rates and policy-rate

indicators, especially under the narrative method, suggesting that markets and monetary authorities may respond more favourably to fiscal packages perceived as credible. Current-expenditure-based consolidations may therefore be associated with easier monetary and financial conditions not because all current spending cuts are inherently more durable, but because some current-expenditure measures, such as pension, transfer or public-sector wage reforms, require substantial political capital and may signal a credible commitment to fiscal adjustment. However, including expenditure shares does not produce a stable monetary effect across the regressions, reinforcing the possibility that part of the monetary-fiscal relationship is endogenous. In other words, interest rates may fall not because monetary policy independently supports consolidation, but because credible consolidation causes monetary authorities and financial markets to react favourably.

A third explanation concerns institutional context, especially the euro area. The euro-area dummies behave differently across identification methods and phases. In the CAPB-based start regression, the 1992-1997 euro-area dummy is positive and highly significant, suggesting that the Maastricht qualification period strongly increased the probability of starting CAPB-identified consolidation. This pattern is not present in the narrative start regression. In the continuation regressions, however, euro-area dummies are often negative, especially after 1998 and, in the narrative results, also during 1992-1997. This suggests that institutional constraints may have encouraged the initial fiscal-balance improvements needed for Maastricht compliance, but did not necessarily ensure that consolidation episodes continued year after year.

This is relevant because the euro area also changed the monetary environment. National monetary policy autonomy declined before disappearing after EMU entry. Therefore, especially for euro-area countries, the measured relationship between interest rates and fiscal consolidation may reflect a combination of domestic fiscal decisions, convergence dynamics, institutional constraints and common monetary policy, rather than a direct national monetary response to fiscal adjustment. Since the sample extends to 2014, covering nine additional years of the euro area compared with Ahrend et al. (2006), the results suggest that institutional context is crucial for identifying monetary-fiscal interactions.

Overall, these findings partially qualify rather than overturn the monetary-fiscal interaction proposed by Ahrend et al. (2006). Monetary conditions appear to affect some aspects of fiscal consolidation, particularly the probability that an episode starts and selected debt-dynamic outcomes. However, these effects are not sufficiently stable across specifications, outcomes and identification methods to support a direct, stable or generalizable monetary policy channel across the full consolidation process. The main implication is therefore that fiscal consolidation remains primarily conditioned by fiscal, macroeconomic and institutional factors, with monetary policy playing a secondary and highly context-dependent role. This is also the central methodological conclusion: if consolidation is identified through observed CAPB improvements, monetary conditions may appear more relevant because they influence the macroeconomic environment and the fiscal balance itself. If consolidation is identified through exogenous policy actions, the monetary channel becomes weaker and less consistent. In this sense, incorrect identification of exogenous consolidation episodes can affect the measured interaction between monetary and fiscal policy by making monetary accommodation appear more important than it is.

5.5 Limitations of the Study

Clearly, the estimated interaction between monetary and fiscal policy appears highly sample-dependent. The CAPB-based results are only weakly consistent with those of Ahrend et al. (2006), despite using a closely related specification, suggesting that changes in country coverage and time period can materially affect the identified role of monetary variables. This is reinforced by the narrative-based results, which produce even more contrasting conclusions: while fiscal need, macroeconomic conditions and adjustment composition show relatively stable effects across samples, monetary variables do not. Although the narrative approach is, in principle, better suited to identifying exogenous fiscal policy actions, the results obtained here may still be constrained by the specific country coverage, time period and number of usable episodes. A larger or differently constructed narrative dataset could therefore produce more reliable and conclusive evidence

on the role of monetary policy. Consequently, the findings should not be interpreted as fully generalizable, but rather as evidence that measuring the influence of monetary conditions is conditional on the sample and episode-identification method considered.

A second limitation concerns the size and structure of the estimation samples. Although the full dataset covers 16 countries between 1978 and 2014, several regressions are estimated on much smaller subsets once the analysis is restricted to ongoing episodes, completed episodes, or episodes that can be classified as seriously pursued. This issue is particularly relevant for the narrative-based success regressions, where the sample is often very small. For example, the narrative regression on whether an episode is seriously pursued contains only 13 observations, and several specifications report pseudo R^2 values equal to 1.00. This suggests that some estimates may be driven by limited variation or near-perfect prediction rather than by a stable underlying relationship. As a result, the absence or presence of significance for monetary variables in these regressions should be interpreted cautiously.

Further, when focusing on the narrative dataset, an additional timing limitation arises. As specified in the construction of the database by Alesina et al. (IGIER, 2026), austerity measures are grouped by the year in which they were introduced rather than by the year of effective implementation. This makes it harder to match exogenous fiscal decisions to yearly macroeconomic and monetary conditions, since the budgetary effect of a measure is mechanically observed only when it is implemented, while households, firms, financial markets and monetary authorities may react already at the time of announcement. Anticipation effects can therefore move output, interest rates and other macroeconomic variables before the fiscal measure is fully implemented. This timing mismatch may partly explain why the narrative-based results differ from the CAPB-based results: the narrative method is closer to policy intent and announcement, while the CAPB method is closer to observed fiscal outcomes and implementation.

A further important limitation concerns the monetary variables themselves. This is central to the interpretation of the results because the analysis uses interest-rate movements as proxies for monetary conditions, even though interest rates do not capture

monetary policy alone. They may also reflect financial-market expectations, inflation developments, sovereign risk premia and the perceived credibility of fiscal consolidation. This is particularly relevant for long-term rates, which may fall because consolidation is perceived as credible, rather than because monetary conditions independently support consolidation. As a result, an observed relationship between lower interest rates and fiscal success may partly reflect the response of financial markets to credible fiscal adjustment, rather than a causal effect of monetary accommodation on consolidation outcomes. Although the analysis includes lagged monetary variables and controls for the composition of fiscal adjustment, reverse causality cannot be fully eliminated. Therefore, the results should be interpreted as conditional associations rather than definitive causal estimates.

Finally, the annual frequency of the data limits the ability to identify the precise timing of monetary-fiscal interactions. Monetary policy can react within months, while fiscal announcements, implementation and market responses may occur at different points within the same year. This is especially relevant in the euro-area context, where the transition to EMU changed the meaning of national interest rates and reduced national monetary-policy autonomy. Overall, these limitations mean that the analysis provides useful evidence on whether monetary conditions are associated with fiscal consolidation, but it cannot fully isolate the causal mechanism through which monetary policy may affect the start, continuation or success of consolidation episodes.

5.6 Directions for Future Research

Future research should therefore focus on improving both the identification of fiscal consolidation episodes and the measurement of surrounding monetary conditions. A first extension would be to use a larger and more detailed narrative dataset, ideally covering more countries and distinguishing clearly between the announcement, legislative approval and effective implementation of fiscal measures, so that monetary conditions can be matched more precisely to the relevant stage of each adjustment. Second, higher-frequency data could be used to examine whether interest rates and financial markets

respond to fiscal news within the year, helping to separate genuine monetary support from reverse causality driven by credibility effects. Finally, future work could distinguish more explicitly between different institutional regimes, especially pre- and post-EMU euro-area countries, and between conventional interest-rate policy and unconventional monetary policy, which has become increasingly relevant following the global financial crisis. This would allow a more precise assessment of whether monetary policy affects fiscal consolidation directly, or whether its apparent role depends mainly on broader macroeconomic conditions, market confidence and institutional constraints.

6 Conclusion

This paper set out to investigate how monetary conditions affect fiscal consolidation, and whether the estimated relationship changes when consolidation episodes are identified through narratively identified exogenous fiscal measures rather than through observed changes in the cyclically-adjusted primary balance. The underlying motivation was therefore both substantive and methodological: to assess whether monetary policy plays an independent role in the start, continuation and success of fiscal consolidation, but also whether an incorrect identification of exogenous consolidation episodes can distort the measured interaction between monetary and fiscal policy.

Overall, the results do not support a stable or systematic monetary-policy channel. Monetary conditions appear to matter in some cases, especially at the start of consolidation episodes, but their effects are not consistent across specifications, outcomes or identification methods. In the CAPB-based regressions, there is some evidence that easier-than-rule monetary conditions are associated with the beginning of consolidation, and that falling short-term rates improve selected debt-dynamic outcomes. However, these results do not extend consistently to continuation, cumulative adjustment, episode length or durability after the episode. Under narrative identification, the evidence is even weaker: short-rate easing is associated with the start of exogenous consolidation episodes, but monetary variables are generally insignificant for continuation and for most measures

of success.

This means that the conclusions reached by Ahrend et al. (2006) are only partially replicated. Easier monetary conditions may help create a more favourable environment for the introduction of fiscal adjustment, and in some cases may reduce the macroeconomic cost of consolidation. However, the evidence in this paper does not show that monetary accommodation systematically makes consolidation more likely to be continued or successful. Instead, the results suggest that fiscal consolidation remains primarily explained by the initial “need for adjustment”, weak macroeconomic conditions, institutional constraints and the credibility and composition of the fiscal package.

The comparison between CAPB-based and narrative-based identification is crucial for interpreting this result. When consolidation episodes are defined through improvements in the CAPB, the estimated interaction with monetary policy is more likely to capture changes in macroeconomic conditions, revenues, interest payments or market confidence that are only partly related to discretionary fiscal action. By contrast, the narrative approach is closer to policy intent, but produces a much less systematic role for monetary variables. Therefore, relying on outcome-based identification can make monetary policy appear more important for fiscal consolidation than it is when exogenous policy decisions are identified more directly.

The main implication is that the measurement of monetary-fiscal interactions depends strongly on the way fiscal consolidation itself is identified. Erroneously treating endogenous improvements in the fiscal balance as exogenous consolidation episodes can generate spurious associations between monetary easing and fiscal success. Although the limitations of the data and the annual frequency of the analysis prevent definitive causal claims, the absence of a stable monetary effect across identification methods suggests that monetary policy should be understood as a contextual factor rather than as an independent and systematic driver of fiscal consolidation. In conclusion, what appears more robust is not the direct role of monetary accommodation, but the importance of fiscal pressure, macroeconomic conditions and credible adjustment design in determining whether consolidation episodes begin, continue and succeed.

Ultimately, before monetary policy can be said to support fiscal consolidation, fiscal consolidation itself must first be correctly identified.

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A Appendix Tables

Table A.1.1: Probability that a consolidation episode is started (Narrative-Based Episode Identification)

Type of regression: Left-hand-side variable	Regression results					
	Pooled probit regressions					
	Binary variable indicating exogenous start of an adjustment episode					
Explanatory variables	1	2	3	4	5	6
Need for adjustment (lagged)	0.01	0.01 *	0.01 *	0.01 *	0.01 *	0.01
Output gap (lagged)	-0.01					
Change in real exchange rate (lagged)	-0.01 ***	-0.01 **	-0.01 **	-0.01 ***	-0.01 ***	-0.01 ***
Euro-area countries, 1978–1991	0.02	-0.03	-0.03	-0.02	-0.02	0.03
Euro-area countries, 1992–1997	0.07	0.05	0.05	0.07	0.07	0.10
Euro-area countries, 1998 onward	0.07	0.05	0.05	0.05	0.05	0.07
Nominal short-term interest rate (lagged change)		-0.03 **				
Real short-term interest rate (lagged change)			-0.03 **			
Nominal long-term interest rate (lagged change)				-0.00		
Real long-term interest rate (lagged change)					-0.01	
Short-term interest rate relative to Taylor rule (lagged change) (constant not reported)						0.02
Observations	251	256	256	256	256	247
Pseudo R ²	0.09	0.10	0.10	0.07	0.07	0.10

Reported coefficients are average marginal effects.
 * significant at 10%; ** significant at 5%; *** significant at 1%.

Table A.1.2: Probability that a consolidation episode is started (CAPB-Based Episode Identification)

Type of regression: Left-hand-side variable	Regression results					
	Pooled probit regressions					
	Binary variable indicating exogenous start of an adjustment episode					
Explanatory variables	1	2	3	4	5	6
Need for adjustment (lagged)	0.02 ***	0.02 ***	0.02 ***	0.02 ***	0.02 ***	0.02 ***
Output gap (lagged)	-0.01					
Change in real exchange rate (lagged)	-0.01	-0.01	-0.01	-0.01	-0.01	-0.00
Euro-area countries, 1978–1991	-0.02	-0.03	-0.03	-0.03	-0.03	-0.04
Euro-area countries, 1992–1997	0.30 ***	0.34 ***	0.34 ***	0.34 ***	0.34 ***	0.33 ***
Euro-area countries, 1998 onward	-0.04	-0.04	-0.04	-0.05	-0.05	-0.04
Nominal short-term interest rate (lagged change)		-0.00				
Real short-term interest rate (lagged change)			-0.00			
Nominal long-term interest rate (lagged change)				0.01		
Real long-term interest rate (lagged change)					0.01	
Short-term interest rate relative to Taylor rule (lagged change) (constant not reported)						-0.06 **
Observations	230	233	233	233	233	224
Pseudo R ²	0.16	0.16	0.16	0.16	0.16	0.19

Reported coefficients are average marginal effects.
 * significant at 10%; ** significant at 5%; *** significant at 1%.

Table A.2.1: Probability that adjustment is continued from previous year (Narrative-Based Episode Identification)

Type of regression: Left-hand-side variable	Regression results											
	Pooled probit regressions											
	Binary variable indicating continuation of an ongoing adjustment episode											
Explanatory variables	1	2	3	4	5	6	7	8	9	10	11	12
Need for adjustment (lagged)	0.02 **	0.02 *	0.02 *	0.02 *	0.02 *	0.02 **	0.02 *	0.02 *	0.02	0.02	0.02 *	0.02 *
Output gap (lagged)	-0.02	-0.03	-0.03	-0.03	-0.03	-0.03	-0.02	-0.02	-0.03	-0.03	-0.02	-0.03
Real exchange rate relative to historical average (lagged)	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	0.01	0.01
Euro-area countries, 1978–1991	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00
Euro-area countries, 1992–1997	-0.25 **	-0.22 *	-0.22 *	-0.24 *	-0.24 *	-0.29 ***	-0.21 **	-0.21 **	-0.18 *	-0.18 *	-0.23 **	-0.25 **
Euro-area countries, 1998 onward	-0.16	-0.17 *	-0.17 *	-0.17 *	-0.17 *	-0.19 *	-0.15 *	-0.15 *	-0.16 **	-0.16 **	-0.15 *	-0.17 **
Nominal short-term interest rate (lagged change)		0.06					0.01					
Real short-term interest rate (lagged change)			0.06					0.01				
Nominal long-term interest rate (lagged change)				0.05					0.06			
Real long-term interest rate (lagged change)					0.05					0.06		
Short-term interest rate relative to Taylor rule (lagged change)						-0.05					-0.02	
Share of expenditure cuts in total adjustment (lagged)							0.04	0.04	0.04	0.04	0.04	
Share of capital-expenditure cuts in total adjustment (lagged) (constant not reported)												-0.01
Observations	101	101	101	101	101	99	85	85	85	85	84	89
Pseudo R2	0.05	0.06	0.06	0.05	0.05	0.06	0.05	0.05	0.06	0.06	0.05	0.06

Reported coefficients are average marginal effects.
* significant at 10%; ** significant at 5%; *** significant at 1%.

Table A.2.2: Probability that adjustment is continued from previous year (CAPB-Based Episode Identification)

Type of regression: Left-hand-side variable	Regression results											
	Pooled probit regressions											
	Binary variable indicating continuation of an ongoing adjustment episode											
Explanatory variables	1	2	3	4	5	6	7	8	9	10	11	12
Need for adjustment (lagged)	0.03 **	0.03	0.03	0.03 *	0.03 *	0.03 *	0.03 **	0.03 **	0.03 ***	0.03 ***	0.03 ***	0.03 *
Output gap (lagged)	-0.03 *	-0.04 **	-0.04 **	-0.03	-0.03	-0.03 *	-0.03 **	-0.03 **	-0.03 *	-0.03 *	-0.03 *	-0.03 *
Real exchange rate relative to historical average (lagged)	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00
Euro-area countries, 1978-1991	-0.15	-0.10	-0.10	-0.18	-0.18	-0.10	-0.11	-0.11	-0.18	-0.18	-0.10	-0.14
Euro-area countries, 1992-1997	-0.07	-0.00	-0.00	-0.06	-0.06	-0.04	-0.02	-0.02	-0.07	-0.07	-0.04	-0.05
Euro-area countries, 1998 onward	-0.20 *	-0.20 *	-0.20 *	-0.19 *	-0.19 *	-0.20 *	-0.18	-0.18	-0.18	-0.18	-0.18	-0.20 *
Nominal short-term interest rate (lagged change)		0.06					0.05					
Real short-term interest rate (lagged change)			0.05					0.05				
Nominal long-term interest rate (lagged change)				0.02					0.02			
Real long-term interest rate (lagged change)					0.02					0.02		
Short-term interest rate relative to Taylor rule (lagged change)						0.03					0.03	
Share of expenditure cuts in total adjustment (lagged)							0.17	0.17	0.19	0.19	0.19	
Share of capital-expenditure cuts in total adjustment (lagged) (constant not reported)												0.11
Observations	122	122	122	122	122	122	122	122	122	122	122	122
Pseudo R2	0.09	0.10	0.10	0.09	0.09	0.09	0.11	0.11	0.10	0.10	0.10	0.09

Reported coefficients are average marginal effects.
 * significant at 10%; ** significant at 5%; *** significant at 1%.

Table A.3.1: Size of yearly fiscal adjustment (Narrative-Based Episode Identification)

Type of regression: Left-hand-side variable	Regression results															
	Fixed-effect panel regressions Change in the primary adjusted fiscal balance (by year)															Pooled OLS regressions
Explanatory variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Need for adjustment (lagged)	0.36***	0.37***	0.35***	0.35***	0.32***	0.32***	0.36***	0.36**	0.38**	0.36***	0.36**	0.32**	0.32**	0.36**	0.16**	0.16**
Output gap (lagged)	-0.09	-0.09	-0.11*	-0.11*	-0.13	-0.13	-0.08	-0.03	-0.05	-0.04	-0.04	-0.09	-0.09	-0.03	-0.15***	-0.15***
Real exchange rate relative to historical average (lagged)		0.02														
Euro-area countries, excluding 1992–1997															-0.23	
Euro-area countries, 1978–1991																
Euro-area countries, 1992–1997															-0.30	-0.30
Euro-area countries, 1998 onward															-0.23	
Nominal short-term interest rate (lagged change)			0.13*							0.08						
Real short-term interest rate (lagged change)				0.13*							0.08					
Nominal long-term interest rate (lagged change)				0.39***								0.35**				
Real long-term interest rate (lagged change)						0.39***							0.35**			
Short-term interest rate relative to Taylor rule (lagged change)							0.02							0.08		
Share of expenditure cuts in total adjustment (lagged)								0.50		0.49	0.49	0.37	0.37	0.55		
Share of capital-expenditure cuts in total adjustment (lagged)									1.01							
(Constant not reported)																
Observations	101	101	101	101	101	101	99	85	89	85	85	85	85	84	101	101
Number of groups	16	16	16	16	16	16	16	14	15	14	14	14	14	14	14	14
Adjusted R ²	0.33	0.34	0.34	0.34	0.40	0.40	0.32	0.29	0.33	0.28	0.28	0.34	0.34	0.28	0.14	0.14

* significant at 10%; ** significant at 5%; *** significant at 1%.

Table A.3.2: Size of yearly fiscal adjustment (CAPB-Based Episode Identification)

Type of regression: Left-hand-side variable	Regression results																Pooled OLS regressions		
	Fixed-effect panel regressions Change in the primary adjusted fiscal balance (by year)																		
Explanatory variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16			
Need for adjustment (lagged)	0.25***	0.25***	0.25***	0.25***	0.22***	0.23***	0.26***	0.27***	0.26***	0.27***	0.26***	0.24***	0.24***	0.27***	0.12***	0.12***	0.12***		
Output gap (lagged)	-0.07	-0.07	-0.08	-0.08	-0.10	-0.10	-0.07	-0.06	-0.07	-0.07	-0.07	-0.09*	-0.09*	-0.07	-0.09*	-0.10**			
Real exchange rate relative to historical average (lagged)		-0.00																	
Euro-area countries, excluding 1992–1997															-0.09				
Euro-area countries, 1978–1991																0.46			
Euro-area countries, 1992–1997															-0.12	-0.13			
Euro-area countries, 1998 onward										0.05						-0.11			
Nominal short-term interest rate (lagged change)			0.04																
Real short-term interest rate (lagged change)				0.04							0.05								
Nominal long-term interest rate (lagged change)					0.23*							0.24*							
Real long-term interest rate (lagged change)						0.24*							0.24*						
Short-term interest rate relative to Taylor rule (lagged change)							0.06							0.09					
Share of expenditure cuts in total adjustment (lagged)								0.57		0.57	0.57	0.61*	0.61*	0.60					
Share of capital-expenditure cuts in total adjustment (lagged)									0.17										
(Constant not reported)																			
Observations	122	122	122	122	122	122	122	122	122	122	122	122	122	122	122	122	122		
Number of groups	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16		
Adjusted R2	0.32	0.31	0.31	0.31	0.35	0.35	0.31	0.32	0.31	0.32	0.32	0.36	0.36	0.32	0.13	0.13	0.13		

* significant at 10%; ** significant at 5%; *** significant at 1%.

Table A.4.1: Probability of episode being seriously pursued after initial phase (Narrative-Based Episode Identification)

Type of regression: Left-hand-side variable	Regression results													
	Pooled probit regressions (one observation per adjustment episode)													
	Binary variable indicating whether a consolidation episode was seriously continued after the initial starting phase													
Explanatory variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Need for adjustment (end of starting phase)	-0.02	-0.04 *	-0.00	-0.00	-0.05 **	-0.05 **	-0.07 *	-0.04 *	-0.04 **	-0.00	-0.00	-0.00	-0.00	-0.00 ***
Output gap (end of starting phase)	-0.01	-0.17 ***	-0.00	-0.00 *	-0.19 ***	-0.18 ***	-0.16 ***	-0.09 **	-0.16 ***	-0.00	-0.00	-0.00	-0.00	-0.00
Real exchange rate relative to historical average (end of starting phase)	-0.03													
Euro-area countries, 1978–1991	1.67 ***	1.51 ***	-0.00	-0.00	1.59 ***	1.59 ***	1.73 ***	1.62 ***	1.46 ***	-0.00	-0.00	-0.00	-0.00	-0.00
Euro-area countries, 1992–1997	-0.41	-1.17 ***	-0.00	-0.00	-1.32 ***	-1.32 ***	-1.29 ***	-0.38	-1.18 ***	-0.00	-0.00	-0.00	-0.00	-0.00
Euro-area countries, 1998 onward	0.10	0.11	-0.00	-0.00	0.06	0.06	-0.07	0.14	0.08	-0.00	-0.00	-0.00	-0.00	-0.00
Nominal short-term interest rate (average change during starting phase)			-0.00							-0.00				
Real short-term interest rate (average change during starting phase)				-0.00						-0.00				
Nominal long-term interest rate (average change during starting phase)					-0.14						-0.00			
Real long-term interest rate (average change during starting phase)						-0.14						-0.00		
Short-term interest rate relative to Taylor rule (average change during starting phase)							-0.15							-0.00
Share of expenditure cuts in total adjustment (starting phase)								1.34 **		-0.00	-0.00	-0.00	-0.00	-0.00
Share of capital-expenditure cuts in total adjustment (starting phase)									-0.12					
(Constant not reported)														
Observations	13	13	13	13	13	13	13	13	13	13	13	13	13	13
Pseudo R2	0.60	0.51	1.00	1.00	0.53	0.53	0.56	0.69	0.51	1.00	1.00	1.00	1.00	1.00

Reported coefficients are average marginal effects.
* significant at 10%; ** significant at 5%; *** significant at 1%.

Table A.4.2: Probability of episode being seriously pursued after initial phase (CAPB-Based Episode Identification)

Type of regression: Left-hand-side variable	Regression results													
	Pooled probit regressions (one observation per adjustment episode)													
	Binary variable indicating whether a consolidation episode was seriously continued after the initial starting phase													
Explanatory variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Need for adjustment (end of starting phase)	0.02	0.02	0.02	0.02	0.03	0.03	0.03	0.04	0.02	0.04	0.04	0.05	0.04	0.04
Output gap (end of starting phase)	-0.09***	-0.09***	-0.10***	-0.10***	-0.09***	-0.09***	-0.08***	-0.07***	-0.09***	-0.08***	-0.08***	-0.08***	-0.08***	-0.07***
Real exchange rate relative to historical average (end of starting phase)	-0.01													
Euro-area countries, 1978–1991	0.38***	0.29**	0.34**	0.34**	0.33***	0.33***	0.19	0.28***	0.36**	0.29***	0.29***	0.35***	0.35***	0.16
Euro-area countries, 1992–1997	-0.18	-0.21	-0.13	-0.13	-0.22	-0.22	-0.29	-0.14	-0.17	-0.10	-0.11	-0.17	-0.17	-0.34**
Euro-area countries, 1998 onward	-0.02	-0.05	-0.06	-0.06	-0.04	-0.04	-0.04	-0.01	-0.05	-0.02	-0.02	-0.00	-0.00	-0.02
Nominal short-term interest rate (average change during starting phase)			0.09							0.02				
Real short-term interest rate (average change during starting phase)				0.09							0.02			
Nominal long-term interest rate (average change during starting phase)					-0.02							-0.04		
Real long-term interest rate (average change during starting phase)						-0.02							-0.04	
Short-term interest rate relative to Taylor rule (average change during starting phase)							-0.08							-0.12*
Share of expenditure cuts in total adjustment (starting phase)								0.59***		0.59***	0.59***	0.58***	0.58***	0.55***
Share of capital-expenditure cuts in total adjustment (starting phase)									0.45					
(Constant not reported)														
Observations	54	54	54	54	54	54	54	54	54	54	54	54	54	54
Pseudo R2	0.24	0.22	0.25	0.25	0.22	0.22	0.24	0.44	0.24	0.44	0.44	0.45	0.45	0.49

Reported coefficients are average marginal effects.
* significant at 10%; ** significant at 5%; *** significant at 1%.

Table A.5.1: Size of cumulated adjustment during episode (Narrative-Based Episode Identification)

Type of regression Left-hand-side variable	Regression results												
	Truncated regressions; pooled sample - one observation per adjustment episode Cumulated change in the primary cyclically adjusted fiscal balance over the whole adjustment episode												
Explanatory variables	1	2	3	4	5	6	7	8	9	10	11	12	13
Need for adjustment (year before episode start)	1.27	0.95	0.94	1.06	1.07	0.28	1.18	1.29	0.96	0.95	1.10	1.10	0.23
Output gap (year before episode start)	0.07	-0.31	-0.32	-0.01	-0.01	-1.00***	0.10	0.16	-0.18	-0.18	0.13	0.13	-1.11***
Real exchange rate relative to historical average (year before episode start)	-0.08												
Euro-area countries, 1978–1991	3.54	0.98	1.08	3.99	3.97	1.53	2.85	1.14	1.21	1.31	3.59	3.58	1.46
Euro-area countries, 1992–1997	3.71	0.15	0.18	3.12	3.10	2.07*	1.97	2.22	-0.09	-0.04	2.47	2.45	2.28*
Euro-area countries, 1998 onward	5.78	3.27	3.31	4.32	4.33	1.55	4.96	4.55	3.47	3.51	4.35	4.36	1.33
Nominal short-term interest rate (average change during episode)		-4.46							-3.86*				
Real short-term interest rate (average change during episode)			-4.43							-3.81			
Nominal long-term interest rate (average change during episode)				1.47						1.34		1.32	
Real long-term interest rate (average change during episode)					1.44								
Short-term interest rate relative to Taylor rule (cumulated change during episode)						2.12***							2.20***
Share of expenditure cuts in total adjustment (cumulated over episode)							2.50		2.07	2.04	2.46	2.47	-0.99
Share of capital-expenditure cuts in total adjustment (cumulated over episode)								-6.14					
(Constant not reported)													
Observations	23	23	23	23	23	23	23	23	23	23	23	23	23

Reported coefficients are truncated-regression coefficients.

* significant at 10%; ** significant at 5%; *** significant at 1%.

Table A.5.2: Size of cumulated adjustment during episode (CAPB-Based Episode Identification)

Type of regression Left-hand-side variable	Regression results												
	Truncated regressions; pooled sample - one observation per adjustment episode												
	Cumulated change in the primary cyclically adjusted fiscal balance over the whole adjustment episode												
Explanatory variables	1	2	3	4	5	6	7	8	9	10	11	12	13
Need for adjustment (year before episode start)	0.78 ***	0.86 ***	0.86 ***	0.81 ***	0.81 ***	0.91 ***	0.87 ***	0.85 ***	0.88 ***	0.88 ***	0.85 ***	0.84 ***	0.92 ***
Output gap (year before episode start)	-0.98 **	-0.76	-0.76	-0.80 *	-0.80 *	-1.05	-0.54 *	-0.91 *	-0.44	-0.44	-0.55 *	-0.55 *	-0.76
Real exchange rate relative to historical average (year before episode start)	0.11												
Euro-area countries, 1978–1991	4.56	4.78	4.73	5.16	5.15	5.25	4.84 *	5.65	4.18	4.16	4.89 *	4.88 *	4.91 *
Euro-area countries, 1992–1997	-1.46	-0.95	-1.00	-1.78	-1.80	-0.17	-0.28	-0.21	-1.05	-1.07	-0.88	-0.91	0.07
Euro-area countries, 1998 onward	1.40	2.11	2.10	1.88	1.87	1.85	2.27	2.01	2.40	2.40	2.20	2.19	1.98
Nominal short-term interest rate (average change during episode)		-0.44							-0.89				
Real short-term interest rate (average change during episode)			-0.49							-0.91			
Nominal long-term interest rate (average change during episode)				-1.35							-0.66		
Real long-term interest rate (average change during episode)					-1.37							-0.69	
Short-term interest rate relative to Taylor rule (cumulated change during episode)						0.49							0.52
Share of expenditure cuts in total adjustment (cumulated over episode)							4.86 **		5.17 **	5.16 **	4.55 **	4.52 **	5.68 **
Share of capital-expenditure cuts in total adjustment (cumulated over episode)								5.23 **					
(Constant not reported)													
Observations	52	52	52	52	52	52	52	52	52	52	52	52	52

Reported coefficients are truncated-regression coefficients.
* significant at 10%; ** significant at 5%; *** significant at 1%.

Table A.6.1: Deceleration in the pace of debt accumulation (Narrative-Based Episode Identification)

Type of regression Left-hand-side variable	Regression results												
	Robust OLS regressions; pooled sample - one observation per adjustment episode Difference in debt accumulation before and after the episode												
Explanatory variables	1	2	3	4	5	6	7	8	9	10	11	12	13
Need for adjustment (year before episode start)	0.68	0.85 *	0.85 *	0.80	0.80	0.66	0.17	1.54 *	0.33	0.34	0.45	0.47	0.11
Output gap (year before episode start)	-0.98 *	-1.59 **	-1.59 **	-1.41 **	-1.40 **	-1.56 *	-0.99	-0.38	-1.16	-1.16	-1.06	-1.06	-1.20
Real exchange rate relative to historical average (year before episode start)	-0.35 **												
Euro-area countries, 1978-1991	-0.13	-5.19	-5.16	-6.50	-6.51	-3.58	-2.61	-14.14 ***	-4.45	-4.44	-4.35	-4.49	-2.97
Euro-area countries, 1992-1997	1.80	-3.99	-4.00	-2.80	-2.79	-6.04							
Euro-area countries, 1998 onward	-0.55	-1.71	-1.68	-1.45	-1.49	-2.32	-2.13	-2.97	-1.50	-1.47	-1.00	-0.92	-2.12
Nominal short-term interest rate (average change during episode)		-1.29							-2.78				
Real short-term interest rate (average change during episode)			-1.29							-2.76			
Nominal long-term interest rate (average change during episode)				-3.31							-1.74		
Real long-term interest rate (average change during episode)					-3.31							-1.84	
Short-term interest rate relative to Taylor rule (cumulated change during episode)						-0.07							0.49
Share of expenditure cuts in total adjustment (cumulated over episode)							-1.12		-0.31	-0.30	-0.46	-0.39	-1.73
Share of capital-expenditure cuts in total adjustment (cumulated over episode)								-20.38 **					
(Constant not reported)													
Observations	20	20	20	20	20	20	13	13	13	13	13	13	13
R-squared	0.73	0.64	0.64	0.63	0.63	0.61	0.45	0.67	0.53	0.53	0.45	0.45	0.45

Reported coefficients are robust OLS coefficients.
* significant at 10%; ** significant at 5%; *** significant at 1%.

Table A.6.2: Deceleration in the pace of debt accumulation (CAPB-Based Episode Identification)

Type of regression Left-hand-side variable	Regression results												
	Robust OLS regressions; pooled sample - one observation per adjustment episode Difference in debt accumulation before and after the episode												
Explanatory variables	1	2	3	4	5	6	7	8	9	10	11	12	13
Need for adjustment (year before episode start)	0.14	0.01	0.02	-0.03	-0.03	0.10	0.03	0.02	0.06	0.06	-0.04	-0.04	0.09
Output gap (year before episode start)	-1.06 **	-1.13 ***	-1.13 ***	-1.35 ***	-1.35 ***	-1.05 ***	-1.32 ***	-1.33 ***	-1.09 ***	-1.10 ***	-1.36 ***	-1.36 ***	-1.05 ***
Real exchange rate relative to historical average (year before episode start)	-0.17 *												
Euro-area countries, 1978-1991	2.01	-0.54	-0.57	2.02	2.03	0.07	0.65	0.66	-0.55	-0.58	2.04	2.05	0.05
Euro-area countries, 1992-1997	3.75 *	1.80	1.71	2.92	2.92	3.66 **	3.54 *	3.52 *	1.85	1.76	2.86	2.86	3.63 **
Euro-area countries, 1998 onward	-1.04	-1.31	-1.31	-1.43	-1.41	-1.48	-1.33	-1.34	-1.27	-1.27	-1.44	-1.43	-1.49
Nominal short-term interest rate (average change during episode)		-1.68 **							-1.77 **				
Real short-term interest rate (average change during episode)			-1.71 **							-1.80 **			
Nominal long-term interest rate (average change during episode)				-1.10							-1.13		
Real long-term interest rate (average change during episode)					-1.11							-1.14	
Short-term interest rate relative to Taylor rule (cumulated change during episode)						-0.66							-0.66
Share of expenditure cuts in total adjustment (cumulated over episode)							0.12		0.83	0.84	-0.28	-0.27	-0.13
Share of capital-expenditure cuts in total adjustment (cumulated over episode)								0.12					
(Constant not reported)													
Observations	30	30	30	30	30	30	30	30	30	30	30	30	30
R-squared	0.74	0.74	0.74	0.71	0.71	0.70	0.69	0.69	0.74	0.74	0.71	0.71	0.70

Reported coefficients are robust OLS coefficients.
* significant at 10%; ** significant at 5%; *** significant at 1%.

Table A.7.1: Length of adjustment episode (Narrative-Based Episode Identification)

Type of regression Left-hand-side variable	Regression results												
	Truncated regressions: pooled sample - one observation per adjustment episode Length of adjustment episode (in years)												
Explanatory variables	1	2	3	4	5	6	7	8	9	10	11	12	13
Need for adjustment (year before episode start)	0.66	0.55	0.55	0.70	0.70	0.64	0.02	-0.03	0.02	0.02	-0.01	-0.01	-0.04
Output gap (year before episode start)	-0.96	-1.11	-1.09	-1.22	-1.22	-1.49	-0.01	0.02	-0.01	-0.01	0.01	0.01	-0.46
Real exchange rate relative to historical average (year before episode start)		0.10	0.10	0.15	0.15	0.13	0.04	-0.03	0.05	0.04	0.03	0.03	0.06 *
Euro-area countries, 1978–1991	0.05	1.07	1.09	-2.16	-2.08	-3.22	-0.41	0.78	-0.55	-0.52	0.26	0.29	-1.33
Euro-area countries, 1992–1997	-1.90	-3.00	-2.96	-4.18	-4.04	-4.90	-1.70	-0.16	-1.78	-1.76	-1.19	-1.17	-1.49
Euro-area countries, 1998 onward	-2.98	-3.39	-3.38	-3.44	-3.44	-3.84	-0.88	-0.52	-0.89	-0.89	-1.00	-1.00	-1.19 *
Nominal short-term interest rate (average change during episode)		2.59							-0.18				
Real short-term interest rate (average change during episode)			2.68							-0.15			
Nominal long-term interest rate (average change during episode)				-1.38							0.82		
Real long-term interest rate (average change during episode)					-1.21							0.85	
Short-term interest rate relative to Taylor rule (cumulated change during episode)						1.19							1.02 *
Share of expenditure cuts in total adjustment (cumulated over episode)							1.34		1.37	1.37	1.23	1.22	1.12
Share of capital-expenditure cuts in total adjustment (cumulated over episode)								4.01					
(Constant not reported)													
Observations	37	37	37	37	37	37	28	28	28	28	28	28	28

Reported coefficients are truncated-regression coefficients.
 * significant at 10%; ** significant at 5%; *** significant at 1%.

Table A.7.2: Length of adjustment episode (CAPB-Based Episode Identification)

Type of regression Left-hand-side variable	Regression results												
	Truncated regressions: pooled sample - one observation per adjustment episode Length of adjustment episode (in years)												
Explanatory variables	1	2	3	4	5	6	7	8	9	10	11	12	13
Need for adjustment (year before episode start)	0.48	0.37	0.38	0.22	0.21	0.41	0.42	0.37	0.39	0.39	0.26	0.26	0.42
Output gap (year before episode start)	-0.98	-1.37	-1.36	-0.93	-0.92	-1.64	-0.68	-1.14	-0.85	-0.84	-0.64	-0.64	-1.28
Real exchange rate relative to historical average (year before episode start)		0.10	0.10	0.09	0.09	0.11	0.13	0.03	0.12	0.12	0.11	0.11	0.14
Euro-area countries, 1978–1991	1.84	2.70	2.64	3.09	3.11	1.40	1.11	2.99	1.95	1.90	2.64	2.65	1.53
Euro-area countries, 1992–1997	-4.30	-2.39	-2.52	-7.03	-7.01	-2.64	-3.98	-2.97	-2.60	-2.69	-5.72	-5.72	-2.15
Euro-area countries, 1998 onward	-6.82	-6.93	-6.95	-5.45	-5.45	-5.68	-5.65	-5.57	-5.69	-5.69	-4.66	-4.68	-4.85
Nominal short-term interest rate (average change during episode)		2.30							1.39				
Real short-term interest rate (average change during episode)			2.18							1.29			
Nominal long-term interest rate (average change during episode)				-3.40						-2.52			
Real long-term interest rate (average change during episode)					-3.39						-2.52		
Short-term interest rate relative to Taylor rule (cumulated change during episode)						1.57							1.59
Share of expenditure cuts in total adjustment (cumulated over episode)							5.83		5.28	5.34	3.70	3.69	6.90
Share of capital-expenditure cuts in total adjustment (cumulated over episode)								12.51					
(Constant not reported)													
Observations	52	52	52	52	52	52	52	52	52	52	52	52	52

Reported coefficients are truncated-regression coefficients.
* significant at 10%; ** significant at 5%; *** significant at 1%.

Table A.8.1: Slippage after episode (Narrative-Based Episode Identification)

Dataset and type of regression Left-hand-side variable	Regression results								
	Robust OLS regressions; pooled sample - one observation per adjustment episode Slippage, i.e. deterioration in the primary cyclically adjusted fiscal balance in the two years after the end of the adjustment episode								
Explanatory variable	1	2	3	4	5	6	7	8	9
Need for adjustment (last year of episode)	0.12	0.01	-0.18	-0.20 *	0.11	0.11	0.13	0.13	0.11
Output gap (last year of episode)	0.39	0.60 *	0.39	0.26 *	0.38	0.37	0.39	0.39	0.45
Euro-area countries, 1978–1991	0.81	1.73	1.41 **	1.08 *	0.75	0.74	0.83	0.83	0.19
Euro-area countries, 1992–1997									
Euro-area countries, 1998 onward	1.28	-0.65	0.12	0.71	1.23	1.22	1.33	1.32	0.70
Cumulated change in primary cyclically adjusted fiscal balance (over episode)		-0.71							
Share of expenditure cuts in total adjustment (cumulated over episode)			-0.55						
Share of capital-expenditure cuts in total adjustment (cumulated over episode)				4.88 **					
Nominal short-term interest rate (cumulated change during episode)					0.07				
Real short-term interest rate (cumulated change during episode)						0.08			
Nominal long-term interest rate (cumulated change during episode)							-0.04		
Real long-term interest rate (cumulated change during episode)								-0.03	
Short-term interest rate relative to Taylor rule (cumulated change during episode)									0.33
(Constant not reported)									
Observations	28	26	18	19	28	28	28	28	27
R-squared	0.15	0.40	0.38	0.53	0.15	0.15	0.15	0.15	0.20

Reported coefficients are robust OLS coefficients.
 * significant at 10%; ** significant at 5%; *** significant at 1%.

Table A.8.2: Slippage after episode (CAPB-Based Episode Identification)

Dataset and type of regression Left-hand-side variable	Regression results								
	Robust OLS regressions; pooled sample - one observation per adjustment episode								
	Slippage, i.e. deterioration in the primary cyclically adjusted fiscal balance in the two years after the end of the adjustment episode								
Explanatory variable	1	2	3	4	5	6	7	8	9
Need for adjustment (last year of episode)	-0.00	-0.00	-0.10	-0.01	-0.03	-0.03	-0.03	-0.03	-0.04
Output gap (last year of episode)	0.31 **	0.32 **	0.34 **	0.29 **	0.23	0.23	0.29 **	0.29 **	0.25 *
Euro-area countries, 1978–1991	-0.30	-0.26	-0.86 *	-0.20	-0.25	-0.25	-0.94	-0.95	-0.04
Euro-area countries, 1992–1997	0.23	0.27	0.50	0.20	0.49	0.50	0.33	0.33	0.38
Euro-area countries, 1998 onward	-0.47	-0.45	-0.53	-0.42	-0.40	-0.40	-0.50	-0.50	-0.50
Cumulated change in primary cyclically adjusted fiscal balance (over episode)		0.04							
Share of expenditure cuts in total adjustment (cumulated over episode)			-1.63 *						
Share of capital-expenditure cuts in total adjustment (cumulated over episode)				0.77					
Nominal short-term interest rate (cumulated change during episode)					0.18				
Real short-term interest rate (cumulated change during episode)						0.19			
Nominal long-term interest rate (cumulated change during episode)							0.16		
Real long-term interest rate (cumulated change during episode)								0.16	
Short-term interest rate relative to Taylor rule (cumulated change during episode) (Constant not reported)									0.22 **
Observations	39	39	39	39	39	39	39	39	39
R-squared	0.15	0.15	0.28	0.16	0.18	0.18	0.17	0.17	0.19

Reported coefficients are robust OLS coefficients.
* significant at 10%; ** significant at 5%; *** significant at 1%.